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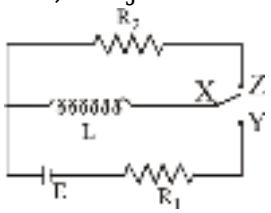
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PHYSICS

1) In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :



(1) $\frac{LE^2}{2R_1^2}$

(2) $\frac{LE^2}{2R_2^2}$

(3) $\frac{LE^2}{2R_1R_2}$

(4) $\frac{LE^2R_2}{2R_1^2}$

2) A coil of inductance 5H is joined to a cell of emf 6V through a resistance 10Ω at time t = 0. The emf across the coil at time t = ln $\sqrt{2}$ s is :-

(1) 3V

(2) 1.5 V

(3) 0.75V

(4) 4.5V

3) The magnetic flux ϕ_B linked with a closed coil depends on time t as $\phi = at^{n^2-1}$, where a is a positive constant and n is any real number. The e.m.f. induced in the coil is e.

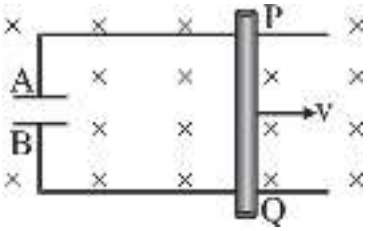
(1) |e| = 0 for n = ± 1

(2) |e| = a for n = $\pm\sqrt{2}$

(3) 1 and 2 both are correct

(4) None of these

4) A conducting rod *PQ* of length L = 1.0 m is moving with a uniform speed *v* = 2 *m/s* in a uniform magnetic field B = 4.0 T directed into the paper. A capacitor of capacity C = 10 μF is connected as shown in figure. Then



(1) $q_A = +80\ \mu C$ and $q_B = -80\ \mu C$

(2) $q_A = -80\ \mu C$ and $q_B = +80\ \mu C$

(3) $q_A = 0 = q_B$

(4) Charge stored in the capacitor increases exponentially with time

5) The output voltage of an ideal transformer, connected to a 240 V a.c. mains is 24 V. When this transformer is used to light a bulb with rating 24V, 24W calculate the current in the primary coil of the circuit.

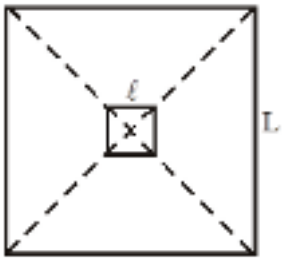
(1) 1A

(2) 2A

(3) 0.2A

(4) 0.1 A

6) A small square loop of wire of side □ is placed inside a large square loop of wire of side (L >> □). The loops are coplanar and their centres coincide. The mutual inductance of the system is :



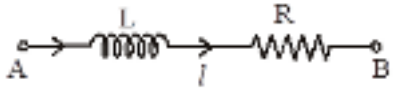
(1) $\frac{2\sqrt{2}\mu_0 L}{\pi \ell}$

(2) $\frac{2\sqrt{2}\mu_0 L^2}{\pi \ell}$

(3) $\frac{2\sqrt{2}\mu_0 \ell}{\pi L}$

(4) $\frac{2\sqrt{2}\mu_0 \ell^2}{\pi L}$

7) In an A.C. sub-circuit as shown in figure, the resistance R = 0.2 Ω. At a certain instant $V_A - V_B$ = 0.5 V, I = 0.5 A, and current is increasing at the rate of $\frac{\Delta I}{\Delta t}$ = 8 A/s. The inductance of the coil is



(1) 0.01 H

(2) 0.02 H

(3) 0.05 H

(4) 0.5 H

8) If magnetic field passing through a coil of area 0.1 m^2 is changing according to the equation B = 10 $\sin\left(\frac{2\pi t}{3}\right)$ tesla. Find the magnitude of induced emf at t = 0.5s

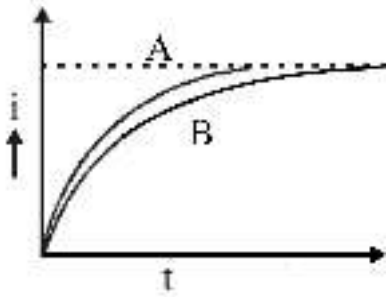
(1) $\frac{\pi}{2}$ volt

(2) $\frac{\pi}{3}$ volt

(3) $\frac{2\pi}{3}$ volt

(4) $\frac{\pi}{4}$ volt

9) A circuit consisting of a constant emf E, a self-inductance L and a resistance R is closed at time t = 0. The relation between the current i in the circuit and the time t is as shown by the curve 'A' in the figure. When one or more of parameters E, R and L are changed, the curve 'B' is obtained. Then it is possible that :-



(1) E and R are kept constant and L is increased

(2) E and R are kept constant and L is decreased

(3) E and L are kept constant and R is decreased

(4) None of these

10) In LR circuit the current increases to one fourth of its maximum value in 4sec than the time constant of circuit is :-

(1) $2\log_e 2$

(2) $\frac{4}{\log_e 2}$

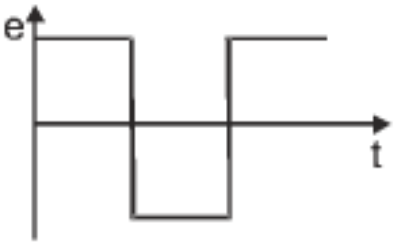
(3) $\frac{4}{\log_e (4/3)}$

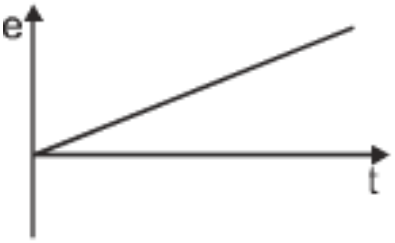
(4) $\frac{\log_e 2}{2}$

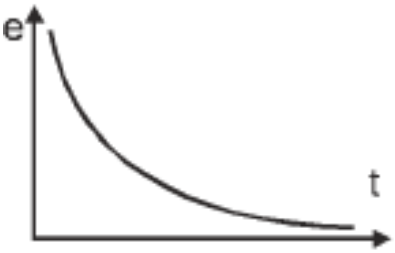
11) A current time curve is shown in the following diagram. This type of current is passed in the primary coil of transformer. The nature of induced emf in the secondary coil will be :–



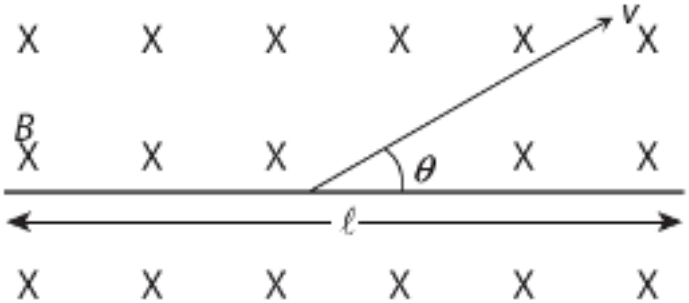
(1)


(2)


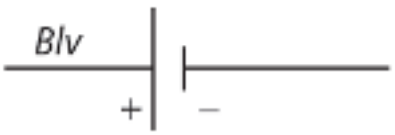
(3)


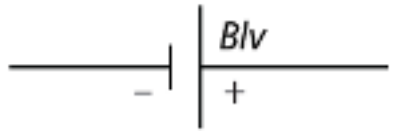
(4)


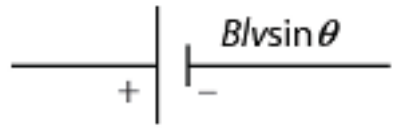
12) A rod of length ℓ is moved with a velocity v in a magnetic field B as shown in the figure, the

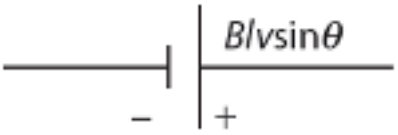


equivalent electrical circuit is -

(1)


(2) 

(3) 

(4) 

13) A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns, giving the output power at 230 V If the current in theprimary of the trnasformer is 5A and its efficiency is 90%, the output current would be

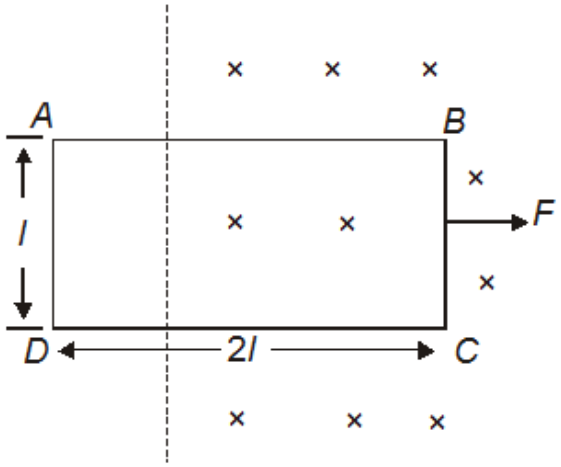
(1) 20 A
(2) 40 A
(3) 45 A
(4) 25 A

14) A metal disc of radius R rotates with an angular velocity ω about an axis perpendicular to its plane passing through its centre in a magnetic field of induction B acting perpendicular to the plane of the disc. The induced e.m.f. between the rim and axis of the disc is

(1) $B\pi R^2\omega$
(2) $BR^2\omega$
(3) $B\pi R^2\omega/2$
(4) $BR^2\omega/2$

15)

A rectangular loop ABCD is being pulled into a uniform magnetic field B with a constant speed (v) with the help of a force F as shown in the figure. The loop has resistance R. Then



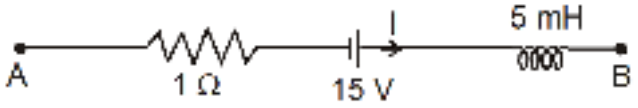
(1) $v = \frac{FR}{Bl}$

(2) $v = \frac{FR}{B^2l^2}$

(3) $v = \frac{FR}{2B^2l^2}$

(4) $v = \frac{FR}{3B^2l^2}$

16) The network shown in figure is part of a complete circuit. If at a certain instant the current (I) is 5 A, and decreasing at a rate of 10^3 A/s, then $V_B - V_A$ =



(1) 15 V

(2) 10 V

(3) 5 V

(4) 20 V

17) A current I = 10 sin (100 πt) ampere, is passed in a coil which induces a maximum emf 5π volt in neighbouring coil. The mutual inductance of two coils is

(1) 5 mH

(2) 10 mH

(3) 15 mH

(4) 25 mH

18) A coil of 10 H inductance and 5 Ω resistance is connected to 5 volt battery in series. The current in ampere in circuit 2 seconds after switched is on :-

(1) e^{-1}

(2) $(1-e^{-1})$

(3) (1–e)

(4) e

19) The magnetic energy stored in a long solenoid of area of cross-section A in a small region of length L is

(1) $\frac{B^2AL}{2\mu_0^2}$

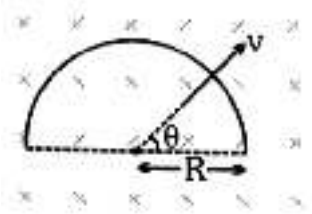
(2) $\frac{AL}{2\mu_0}$

(3) $\frac{1}{2}\mu_0 B^2 AL$

(4) $\frac{B^2AL}{2\mu_0}$

20) A semi circular loop of radius R is placed in a uniform magnetic field as shown. It is pulled with a

constant velocity. The induced emf in the loop is :



(1) $Bv(\pi R)\cos\theta$

(2) $Bv\pi R\sin\theta$

(3) $Bv(2R)\cos\theta$

(4) $Bv(2R)\sin\theta$

21) A wire loop is rotated in uniform magnetic field. The frequency of change of direction of the induced current is :

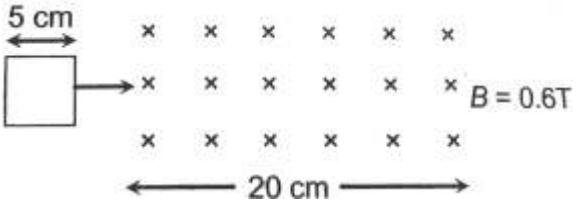
(1) Six times per revolution

(2) Once per revolution

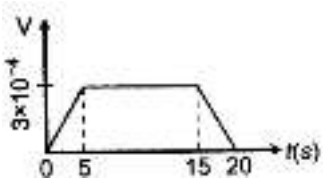
(3) Twice per revolution

(4) Four times per revolution

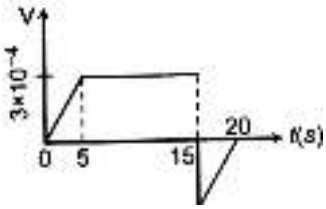
22) A square loop of side 5cm enters a magnetic field with 1cm/s. The front edge enters the magnetic field at t=0, then which graph best depicts emf induced?



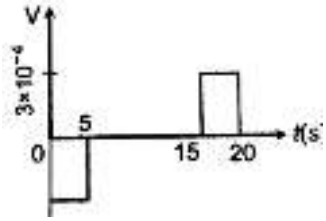
(1)

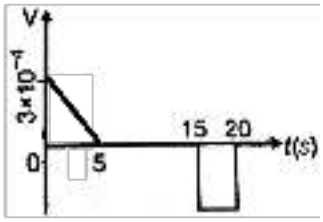


(2)



(3)



(4) 

23) Power factor of an L-R series circuit is 0.6 and that of a C–R series circuit is 0.5. If the element (L, C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance in the L-R circuit to the resistance in the C–R circuit is :-

(1) 6/5

(2) 5/6

(3) $\frac{4}{3\sqrt{3}}$

(4) $\frac{3\sqrt{3}}{4}$

24) The voltage supplied to a circuit is given by $V = V_0t^{3/2}$, where t is time in second. Find the RMS value of voltage for the period t = 0 to t = 1s :-

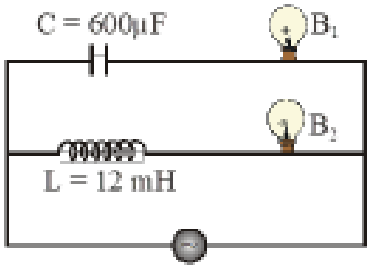
(1) $\frac{V_0}{2}$

(2) V_0

(3) $\frac{3V_0}{2}$

(4) $2V_0$

25) In the circuit shown in figure, if both the bulbs B_1 and B_2 identical then :-



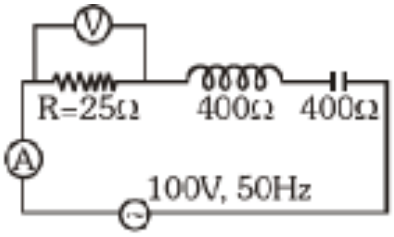
(1) Both bulbs have equal brightness

(2) B_2 will be brighter than B_1

(3) At very high frequency of source brightness of B_1 is more than B_2.

(4) At the very low frequency of source the brightness of B_1 is more than B_2.

26) In given LCR circuit, the voltage across the terminals of a resistance & current will be :



(1) 400V, 2A

(2) 800V, 2A

(3) 100V, 2A

(4) 100V, 4A

27) An alternating current is given by $i = i_0 + i_1 \sin wt$ then its rms value will be :

(1) $\sqrt{i_0^2 + 0.5i_1^2}$

(2) $\sqrt{i_0^2 + 0.5i_0^2}$

(3) 0

(4) $i_0/\sqrt{2}$

28) An inductor of inductance L and resistor of resistance R are joined in series and connected by a source of frequency ω. Power dissipated in the circuit is :

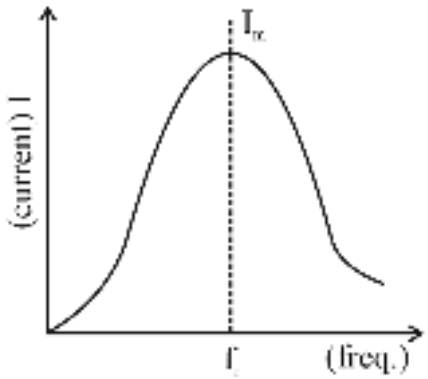
(1) $\frac{\left(R^2 + \omega^2 L^2\right)}{V}$

(2) $\frac{V^2 R}{\left(R^2 + \omega^2 L^2\right)}$

(3) $\frac{V}{\left(R^2 + \omega^2 L^2\right)}$

(4) $\frac{\sqrt{R^2 + \omega^2 L^2}}{V^2}$

29) The graph shows variation of I with f for a series R-L-C network. Keeping L and C constant. If R decreases :



(a) Maximum current (I_m) increases

(b) Sharpness of the graph increases

(c) Quality factor increases
(d) Band width increases

(1) a, b, c
(2) b, c, d
(3) c, d, a
(4) All

30) An a.c. source given by $V = V_m \sin\omega t$ is connected to a pure inductor L in a circuit and I_m is the peak value of the ac current. The instantaneous power supplied to the inductor is :

(1) $-\frac{V_m I_m}{2}\sin(2\omega t)$
(2) $V_m I_m \sin^2(\omega t)$
(3) $-V_m I_m \sin^2(\omega t)$
(4) $\frac{V_m I_m}{2}\sin(2\omega t)$

31) In a transformer the input voltage and input current are 11 kV and 10 A respectively. If the output voltage is 450 V and the output current is 220 A, then efficiency of the transformer is approximately

(1) 9%
(2) 900%
(3) 90%
(4) 80%

32) If angular frequency of an A.C. source used in L-C-R circuit is decreased, then inductive reactance ..... and capacitive reactance.....

(1) Decrease, increase
(2) Increase, increase
(3) Increase, decrease
(4) Decrease, decrease

33) If in a L-R series circuit the power factor is $\frac{1}{2}$ and R = 100Ω, then the value of L is, when AC mains is used

(1) π H
(2) $\frac{\sqrt{3}}{\pi}$ H
(3) $\frac{\pi}{\sqrt{3}}$ H

(4) $\frac{\sqrt{2}}{\pi}$H

34) A resistance R draws power P when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes Z the power drawn will be

(1) P

(2) $P\left(\frac{R}{Z}\right)$

(3) $P\left(\frac{R}{Z}\right)^2$

(4) $\frac{PR^2}{Z}$

35) In L-C-R series circuit, the potential difference across each element is 60 V. Now the value of resistance alone is doubled, then potential drop across R, L and C respectively is

(1) 60 V, 30 V, 30 V

(2) 60 V, 120 V, 120 V

(3) 30 V, 120 V, 120 V

(4) 60 V, 100 V, 80 V

36) When a voltage $v_s = 200\sqrt{2}$ sin (ω t + 15°) is applied to an AC circuit the current in the circuit is found to be i = 2 sin (ω t + π/4) then average power consumed in the circuit is

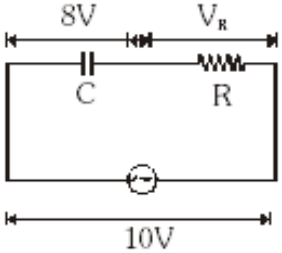
(1) 200 watt

(2) 400 $\sqrt{2}$ watt

(3) 100 $\sqrt{6}$ watt

(4) 200 $\sqrt{2}$ watt

37) In a series CR circuit shown in figure, the applied voltage is 10V and the voltage voltage across capacitor is found to be 8V. then the voltage across R, and the phase difference between current and the applied voltage will respectively be



(1) 6V, $\tan^{-1}\left(\frac{4}{3}\right)$

(2) 3V, $\tan^{-1}\left(\frac{3}{4}\right)$

(3) 6V, $\tan^{-1}\left(\frac{5}{3}\right)$

(4) None

38) The time required for a 50 Hz sinusoidal alternating current to change its value from zero to the r.m.s. value

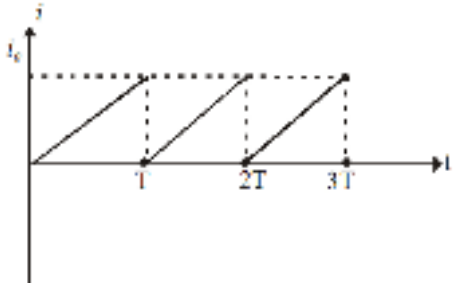
(1) 1.5×10^{-2} s

(2) 2.5×10^{-3} s

(3) 10^{-1} s

(4) 10^{-6} s

39) RMS value is:-



(1) i_0

(2) $\frac{i_0}{\sqrt{3}}$

(3) $\frac{i_0}{\sqrt{6}}$

(4) $\frac{i_0}{\sqrt{2}}$

40) An L-C-R series circuit is at resonance with 10 V each across L, C are R. If the resistance is halved the respective voltages across L, C and R are :-

(1) 10 V, 10 V and 5 V

(2) 10 V, 10 V and 10 V

(3) 20 V, 20 V and 5 V

(4) 20 V, 20 V and 10 V

41) An AC voltage is represented by :-

$E = 575\sqrt{5}\cos(75\pi)t$

How many times will the current become zero in 1 second ?

(1) 75 times

(2) 50 times

(3) 100 times

(4) 575 times

42) The impedance of a R-C circuit is Z_1 for a frequency f and Z_2 for frequency 2f. Then, Z_1/Z_2 is:-

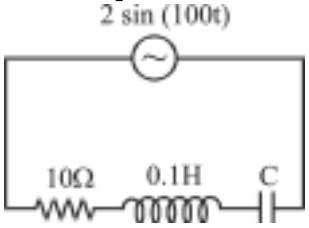
(1) between 1 and 2

(2) 2

(3) between 1/2 and 1

(4) 1/2

43) The power factor of circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to :-



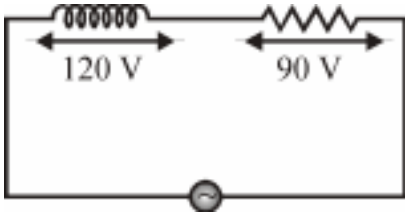
(1) 400 µF

(2) 300 µF

(3) 500 µF

(4) 200 µF

44) If current in the circuit is 3A, then impedance of the circuit shown in the figure is :-



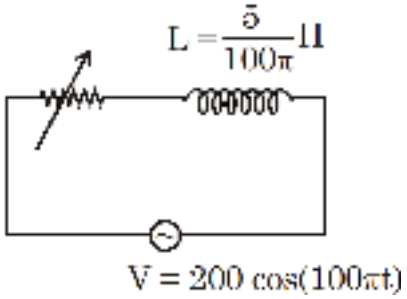
(1) 100 Ω

(2) 50 Ω

(3) 30 Ω

(4) 40 Ω

45) We have attached a variable resistor in an L-R circuit connected to an AC source as shown. At



what value of R will power dissipated in it be maximum ?

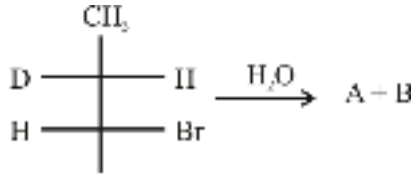
(1) $5\sqrt{2}\Omega$

(2) $\frac{5}{\sqrt{2}}\Omega$

(3) 5Ω

(4) $\frac{5}{2}\Omega$

CHEMISTRY



1) , A and B are:-

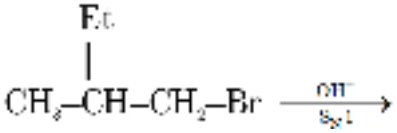
(1) Enantiomers

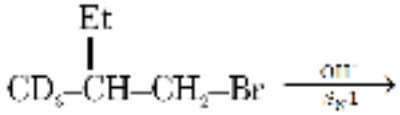
(2) Diastereomers

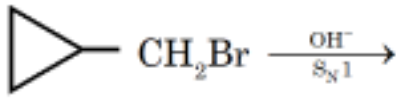
(3) Identical

(4) Position isomers

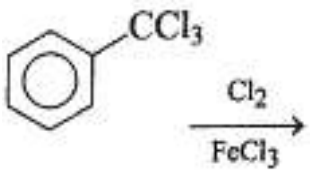
2) In which case recemisation takes place.

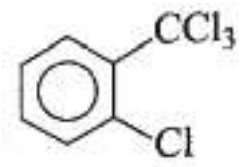
(1) 

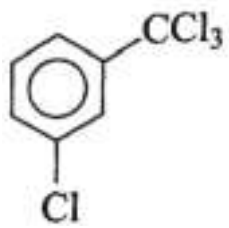
(2) 

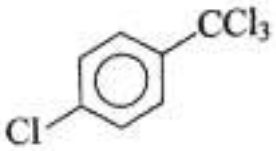
(3) 

(4) All of these

3) Find the major product in the following reaction 

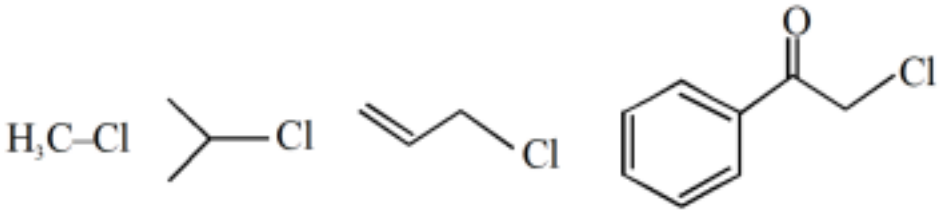
(1) 

(2) 

(3) 

(4) 

4) KI in acetone, undergoes S_N2 reaction with each of P, Q, R and S. The rates of the reaction vary as



- **P** **Q** **R** **S**

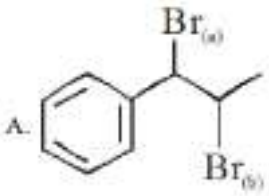
(1) P > Q > R > S

(2) S > P > R > Q

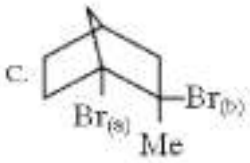
(3) P > R > Q > S

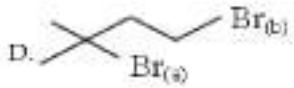
(4) R > P > S > Q

5) Choose the halogen which is most reactive towards S_N1 reaction in the given compounds (A, B, C & D).









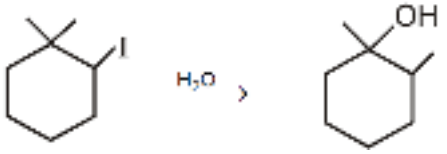
(1) A-$Br_{(b)}$; B-$I_{(b)}$; C-$Br_{(b)}$; D-$Br_{(b)}$

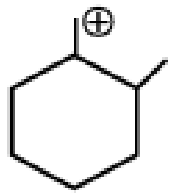
(2) A-$Br_{(a)}$; B-$I_{(a)}$; C-Br(b) ; D-$Br_{(a)}$

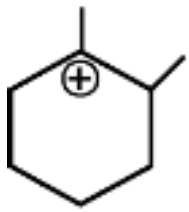
(3) A-$Br_{(a)}$; B-$I_{(a)}$; C-$Br_{(a)}$; D-$Br_{(a)}$

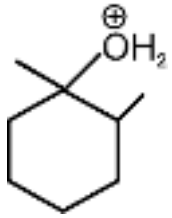
(4) A-$Br_{(b)}$; B-$I_{(a)}$; C-$Br_{(a)}$; D-$Br_{(a)}$

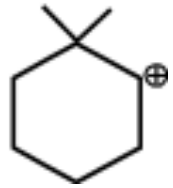
6) Which of the following is not expected to be intermediate of the following reaction ?

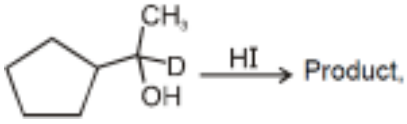


(1) 

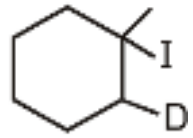
(2) 

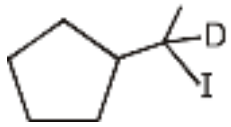
(3) 

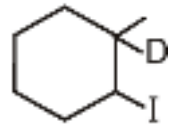
(4) 

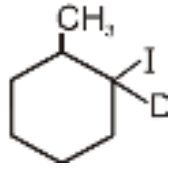
7) 

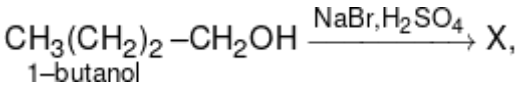
Identify the major product :

(1) 

(2) 

(3) 

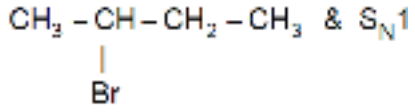
(4) 

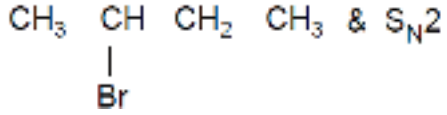
8) 

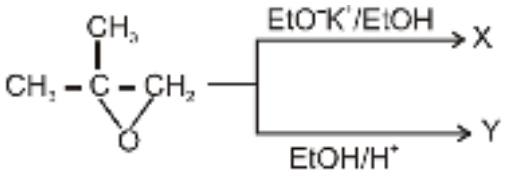
Identify X and the mechanism of the reaction

(1) $CH_3 - CH_2 - CH_2 - CH_2 - Br$ & S_N1

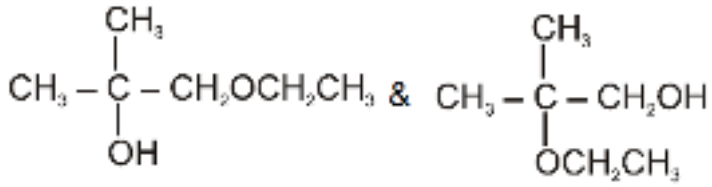
(2) $CH_3 - CH_2 - CH_2 - CH_2 - Br$ & S_N2

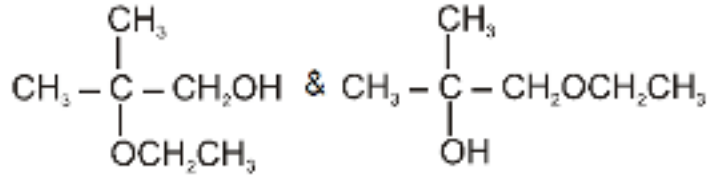
(3) 

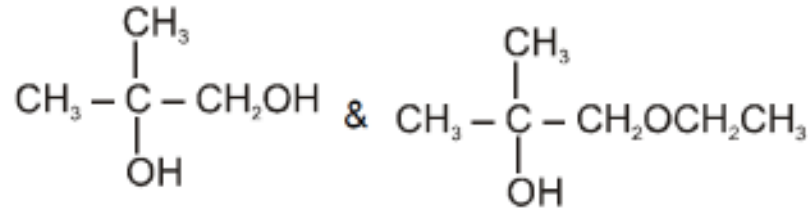
(4) 

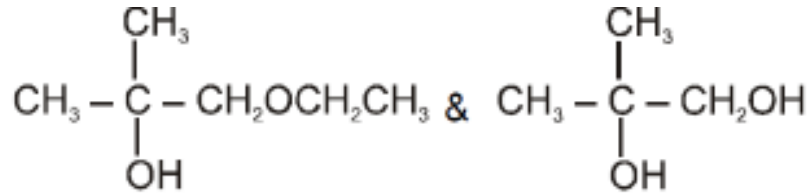
9) 

The product X and Y are respectively :

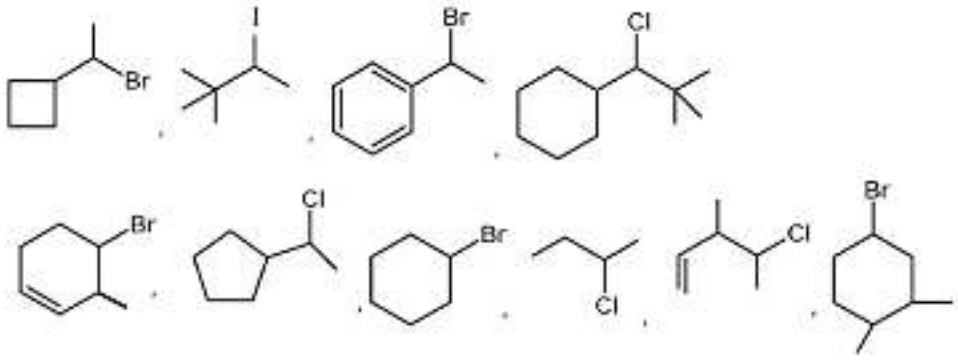
(1) 

(2) 

(3) 

(4) 

10) How many substrates will show rearrangement during S_N1 reaction?



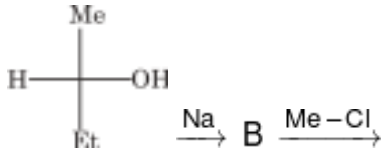
(1) 4

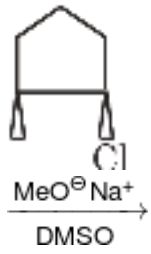
(2) 5

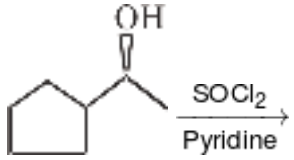
(3) 6

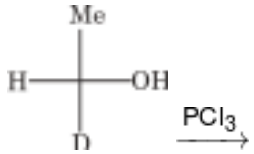
(4) 7

11) In which of the following reaction only retention of configuration take place :

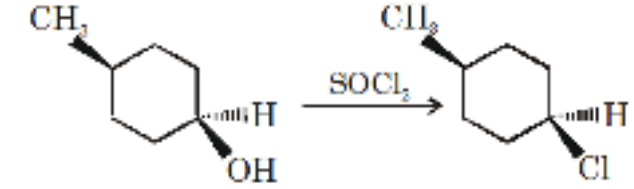
(1) 

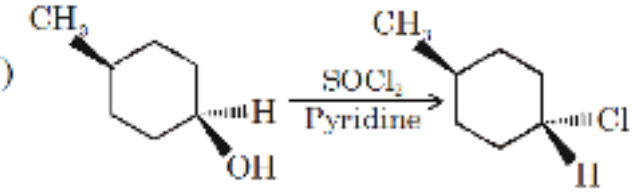
(2) 

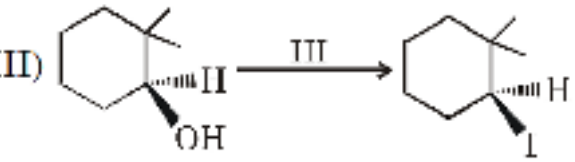
(3) 

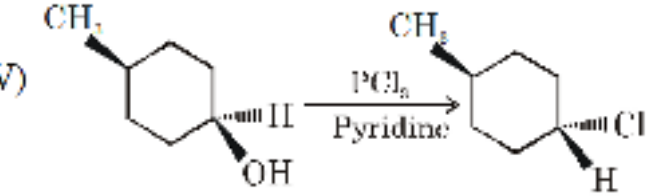
(4) 

12) Which of the following reaction is showing incorrect major product :

(I) 

(II) 

(III) 

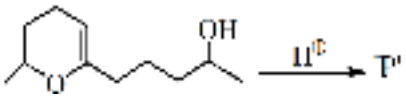
(IV) 

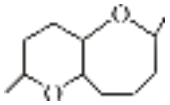
(1) I

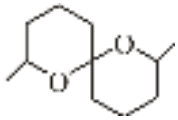
(2) II

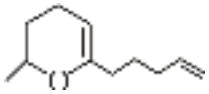
(3) III

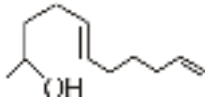
(4) IV

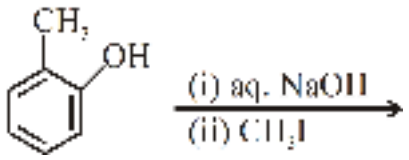
13)  (Major Product)

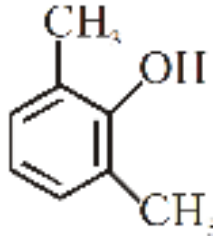
(1) 

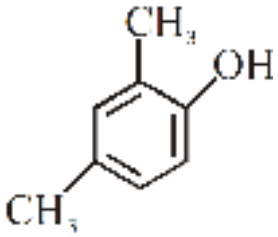
(2) 

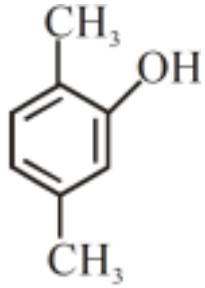
(3) 

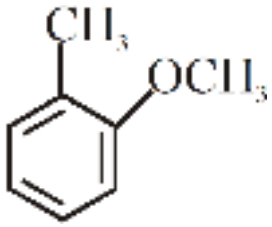
(4) 

14) The major product of the following reaction is: 

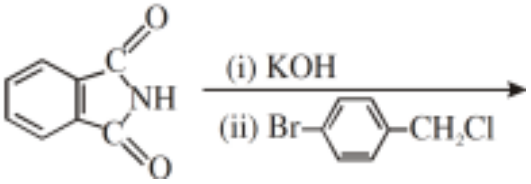
(1) 

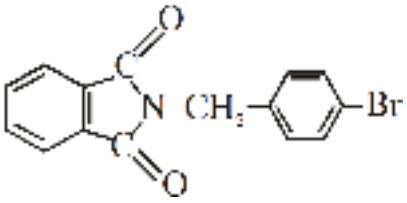
(2) 

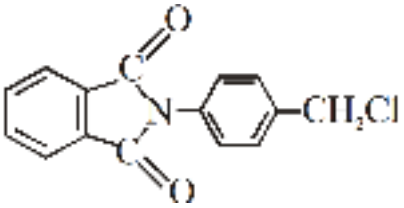
(3) 

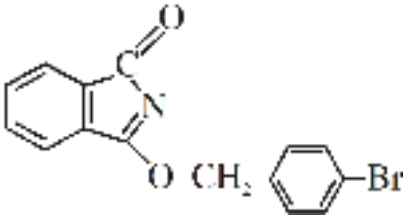
(4) 

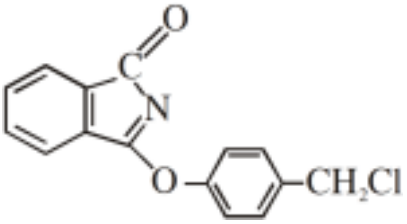
15) The major product of the following reaction is

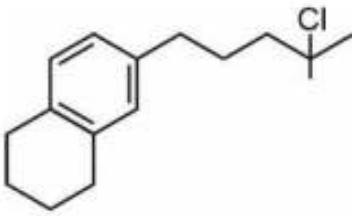


(1) 

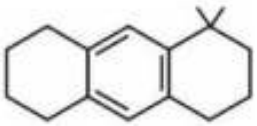
(2) 

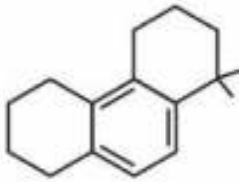
(3) 

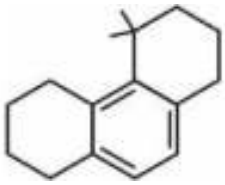
(4) 

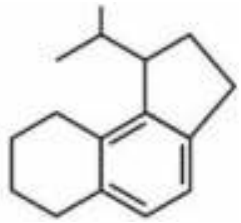
16) 

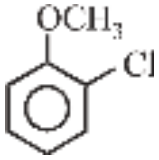
$\xrightarrow[AlCl_3]{\text{Anhydrous}}$ Major product (P). Then product (P) is

(1) 

(2) 

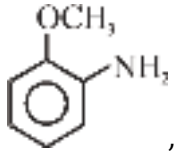
(3) 

(4) 

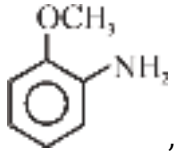
17) 

?

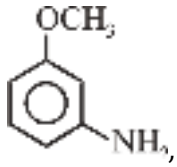
What is major product and type of reaction is ?

(1) 

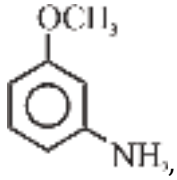
, nucleophilic addition

(2) 

, elimination reaction

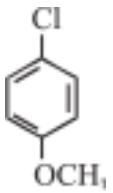
(3) 

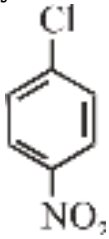
, cine substitution

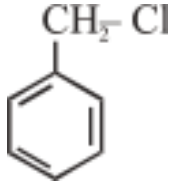
(4) 

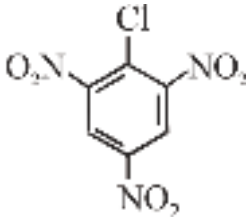
, electrophilic substitution

18) Find the reactivity order towards hydrolysis.

(P) 

(Q) 

(R) 

(S) 

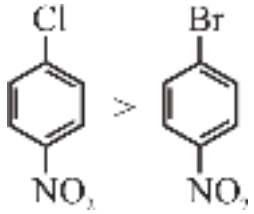
(1) R > S > Q > P

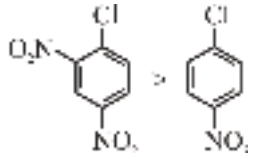
(2) R > P > Q > S

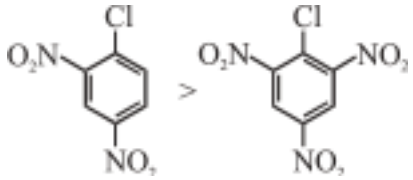
(3) P > Q > R > S

(4) S > Q > P > R

19) Identify correct reactivity for S_NAr reaction

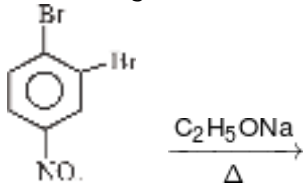
(1) 

(2) 

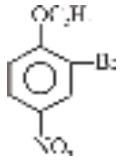
(3) 

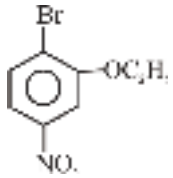
(4) (1) and (2) are correct

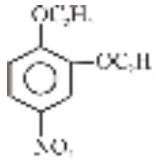
20) In the given reaction :

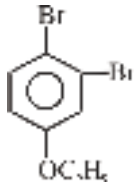


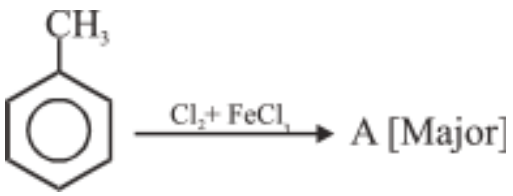
(P), (P) is :-

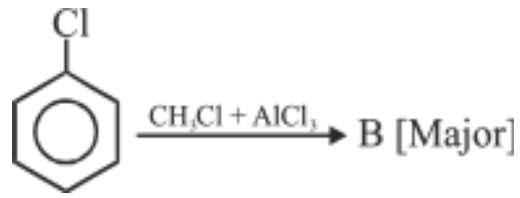
(1) 

(2) 

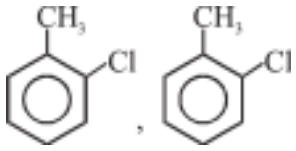
(3) 

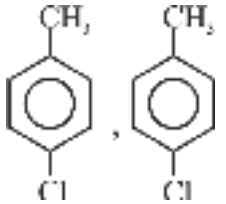
(4) 

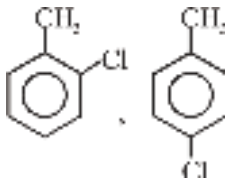
21) 

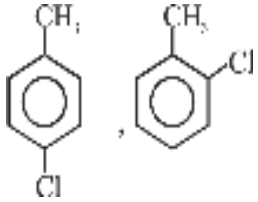


A & B will be respectively :-

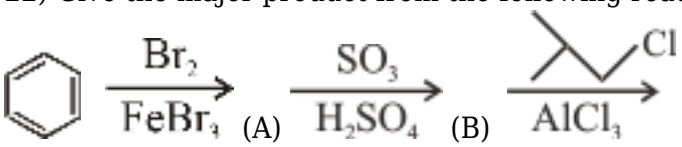
(1) 

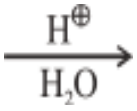
(2) 

(3) 

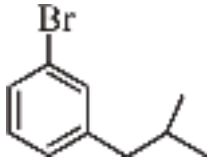
(4) 

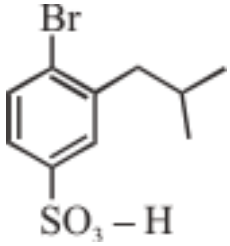
22) Give the major product from the following reaction sequence :-

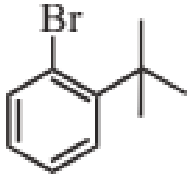


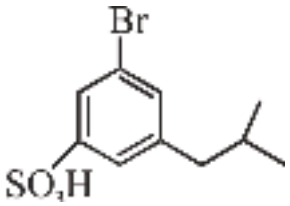


(C) (D) Major product :-

(1) 

(2) 

(3) 

(4) 

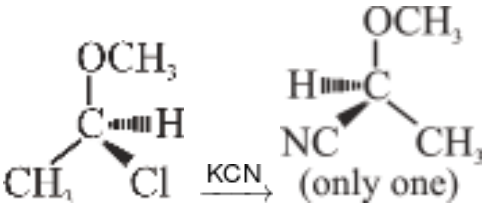
23) An incorrect statement with respect to S_N1 and S_N2 mechanisms for alkyl halide is -

(1) A strong nucleophile in an aprotic solvent increases the rate of S_N2 reaction

(2) In S_N2 reaction back side attack take place generally

(3) Rate of S_N1 reaction increase with increase the concentration of nucleophile

(4) a weak nucleophile and a protic solvent increases the rate or favours S_N1 reaction.

24) 
Correct statement about above reaction is ?

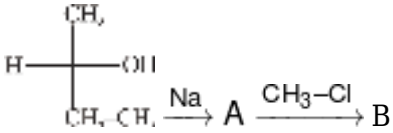
(1) SN^1 with retained configuration

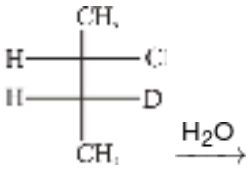
(2) SN^2 with retained configuration

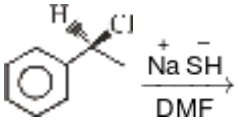
(3) SN^1 with inverson of configuration

(4) SN^2 with inverson of configuration

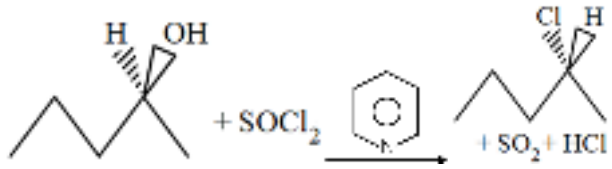
25) In which reaction walden inversion take place?

(1) 

(2) 

(3) 

(4) $CH_3-CH_2-CH_2-CH_3+Cl_2 \xrightarrow{h\nu}$

26) 
Proceed by the mechanism

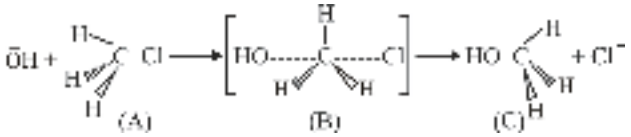
(1) S_{N^1}

(2) S_{N^2}

(3) S_NA_r

(4) ESR

27) Correct statement for



(1) It is unimolecular nucleophilic substitution

(2) B have less energy than A and C

(3) Behaviour of A and C towards plane polarised light are different

(4) Reaction rate is depend on conc. of OH^-

28)  + 1 mole (NaI) $\xrightarrow{\text{Acetone}}$ (A) Major

Find A.

(1) 

(2) 

(3) 

(4) 

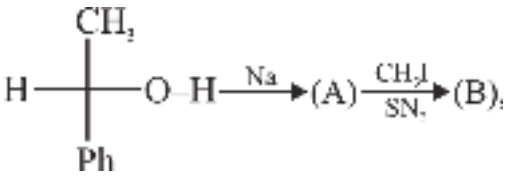
29) The most suitable reagent for the preparation of R-Cl from alcohol is

(1) PCl_5

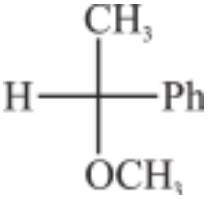
(2) $ZnCl_2$/HCl

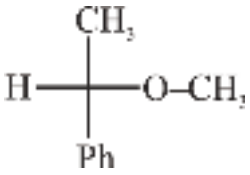
(3) PCl_3

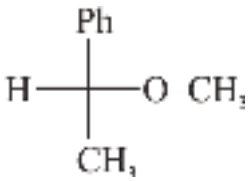
(4) $SOCl_2$/Pyridine

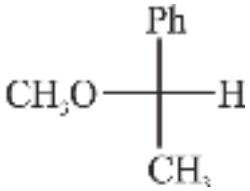
30) 

Product (B) is :-

(1) 

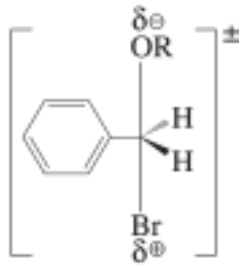
(2) 

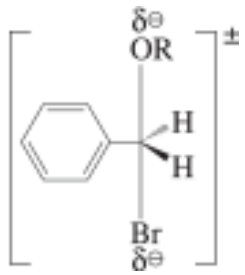
(3) 

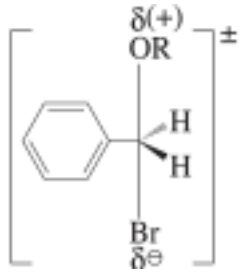
(4) 

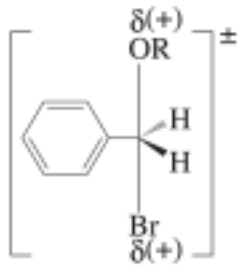
31) Transition state of given S_{N2} is:



(1) 

(2) 

(3) 

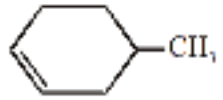
(4) 

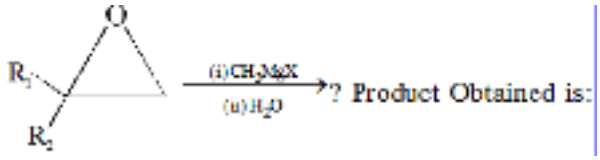
32)  Major product of reaction is :-

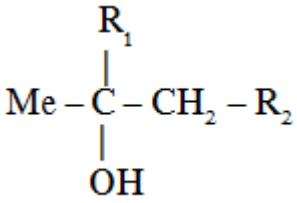
(1) 

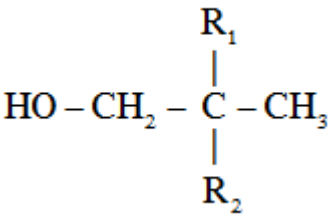
(2) 

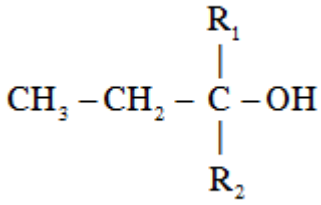
(3) 

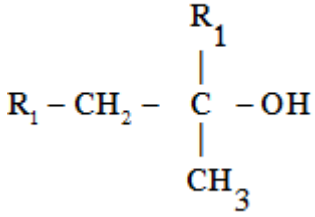
(4) 

33) 

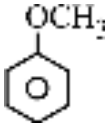
(1) 

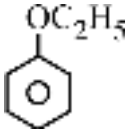
(2) 

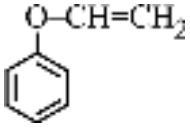
(3) 

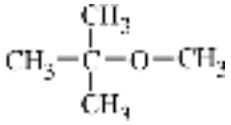
(4) 

34) Which one not obtained by Williamson ether synthesis?

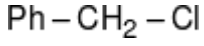
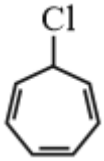
(1) 

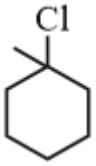
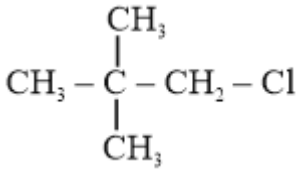
(2) 

(3) 

(4) 

35) Arrange the following compounds in decreasing order of reactivity in S_N1 reaction :-

(a)  (b) 

(c)  (d) 

(1) a > c > b > a

(2) c > d > b > a

(3) a > b > c > d

(4) b > a > c > d

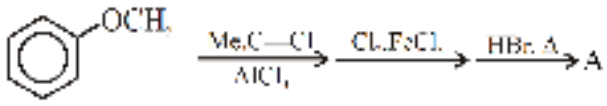
36) Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**. **Assertion (A) :** t-butyl methyl ether is prepared by reaction of t-butyl bromide with methoxide ion. **Reason (R) :** This reaction follows S_N2 mechanism.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

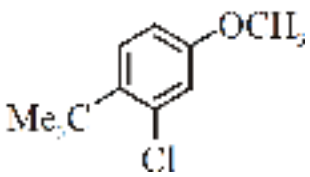
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

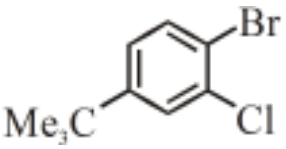
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

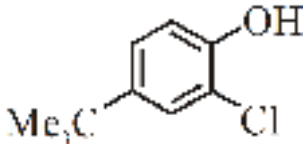
(3) (A) is correct but (R) is not correct.

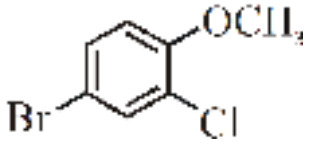
(4) (A) and (R) both are incorrect.

37)  The final product A is :

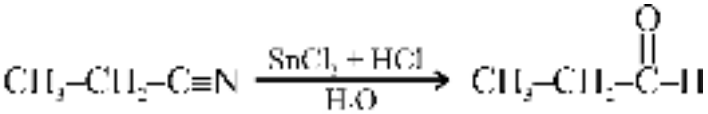
(1) 

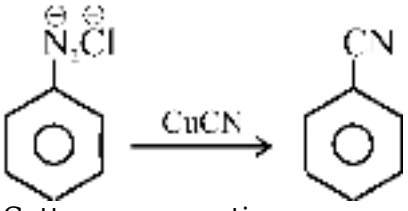
(2) 

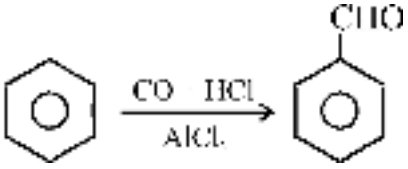
(3) 

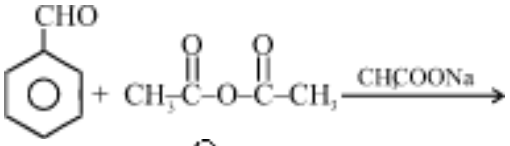
(4) 

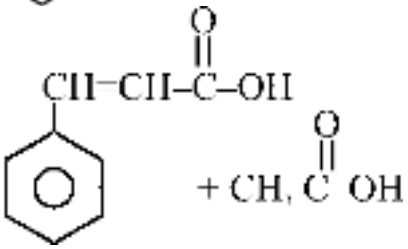
38) Which of the following name reaction is not correct:-

(1) 
Stephen reaction

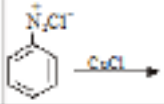
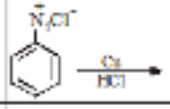
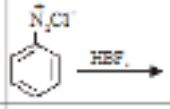
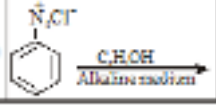
(2) 
Gattermann reaction

(3) 
Gattermann Koch Aldehyde synthesis



(4) 
Perkin's Reaction

39) Match the column I with column II and mark the appropriate choice :-

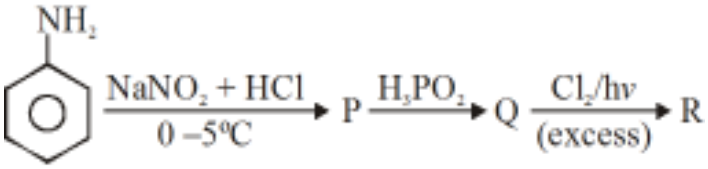
Column I	Column II
(A) 	(i) Coupling reaction
(B) 	(ii) Balz Schiemann reaction
(C) 	(iii) Gattermann reaction
(D) 	(iv) Sandmeyer reaction

(1) (A)→(iv), (B)→(iii), (C)→(ii), (D)→(i)

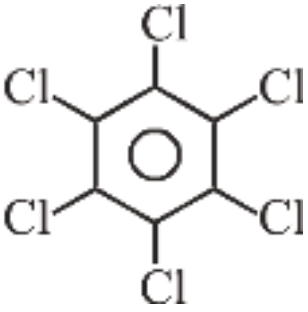
(2) (A)→(iii), (B)→(iv), (C)→(i), (D)→(ii)

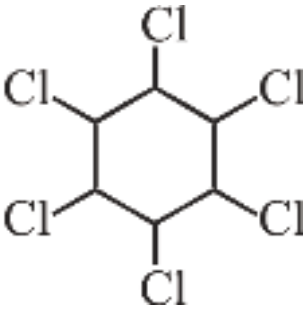
(3) (A)→(ii), (B)→(iii), (C)→(iv), (D)→(i)

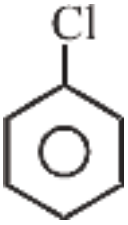
(4) (A)→(i), (B)→(ii), (C)→(iii), (D)→(iv)

40) 

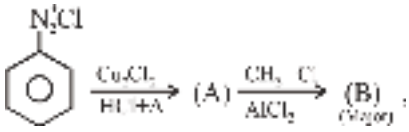
R is :-

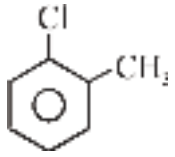
(1) 

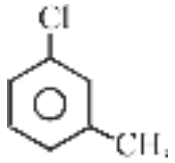
(2) 

(3) 

(4) 

41)  find (B)

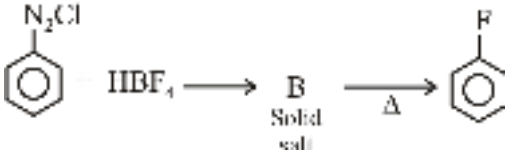
(1) 

(2) 

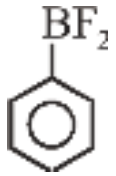
(3) 

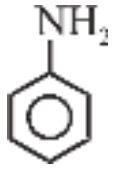
(4) 

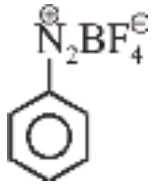
42) Suppose following reaction



The structure of intermediate compound B is :-

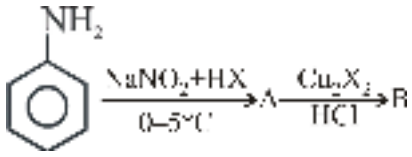
(1) 

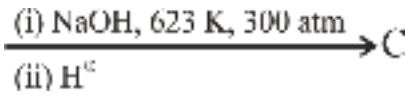
(2) 

(3) 

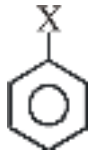
(4) None of these

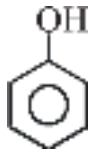
43)



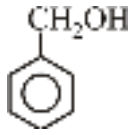


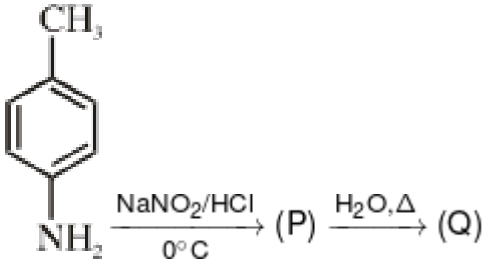
Find product (C) :-

(1) 

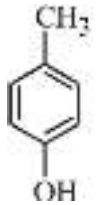
(2) 

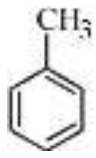
(3) 

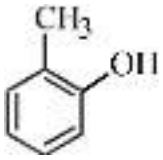
(4) 

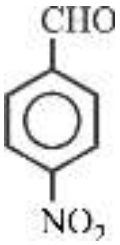
44) 

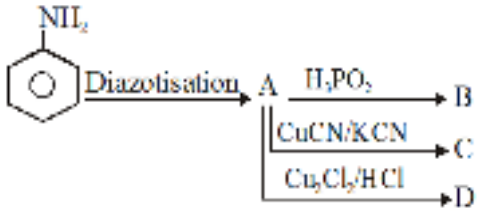
The product 'Q' in the above reaction is

(1) 

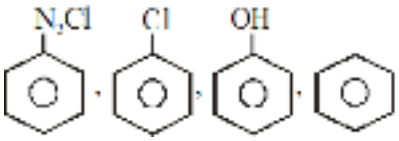
(2) 

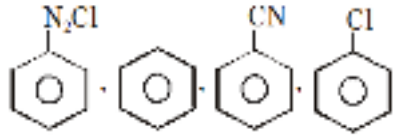
(3) 

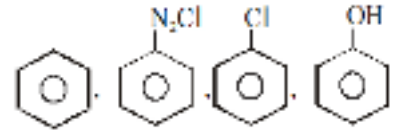
(4) 

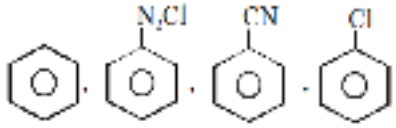
45) 

A, B, C and D are respectively

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Match the column :

Column I	**Column II**
A. ϕ × 174 DNA	(i) 3.3×10^9 bp
B. λ - phage DNA	(ii) 4.6×10^6 bp
C. E. coli DNA	(iii) 48502 bp
D. Haploid content of human DNA	(iv) 5386 bases

(1) A = (iv), B = (iii), C = (ii), D = (i)

(2) A = (i), B = (ii), C = (iii), D = (iv)

(3) A = (ii), B = (iii), C = (iv), D = (i)

(4) A = (i), B = (iv), C = (ii), D = (iii)

2) Which of the following is found in a DNA?

(1) dATP

(2) GMP

(3) dUMP

(4) Deoxyribonucleoside monophosphates

3) if in one of the strands of a ds DNA $\frac{A+G}{T+C}=\frac{3}{5}$ then, what will be the ratio of $\frac{A+G}{T+C}$ in the other strand?

(1) $\frac{5}{3}$

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) $\frac{1}{5}$

4) Which of the following is correct for a dsDNA molecule?

(1) The two chains have antiparallel polarity which means one chain has the polarity 5'→ 3' and other also has 5' → 3' polarity

(2) It is a polymer of deoxyribonucleoside triphosphates

(3) The pitch of helix is 3.4 Å

(4) The backbone is constituted by sugar-phosphate and bases project inside

5) The prokaryotic DNA is nucleoid which is organised in large loops held by:

(1) Positively charged polyamines (positively charged proteins)

(2) Negatively charged proteins

(3) Neutral amino acids

(4) RNA only

6) In a nucleosome, the histone core is made of :

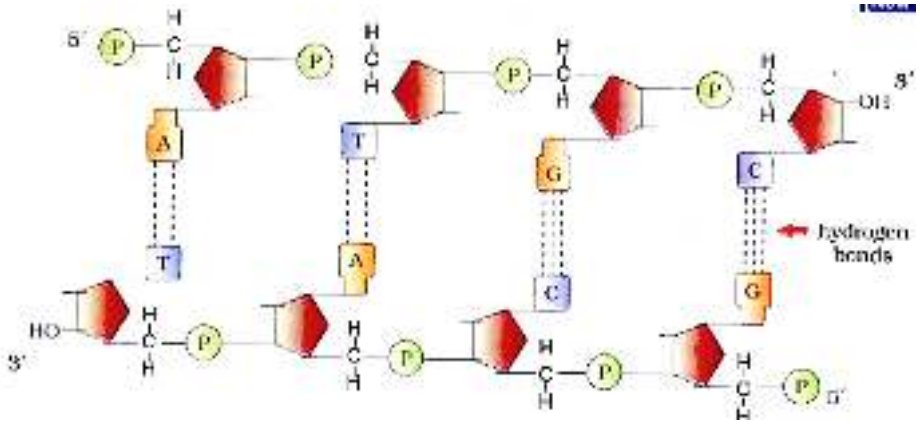
(1) 2(H2A + H2B + H3 + H4)

(2) 2(H1 + H2 + H3 + H4)

(3) 4(H2A + H2B + H3 + H4)

(4) 8(H2A + H2B + H3 + H4)

7) How many phosphodiester bonds (PDB) and N - glycosidic bonds(NGB) are present in a stretch of dsDNA given below:



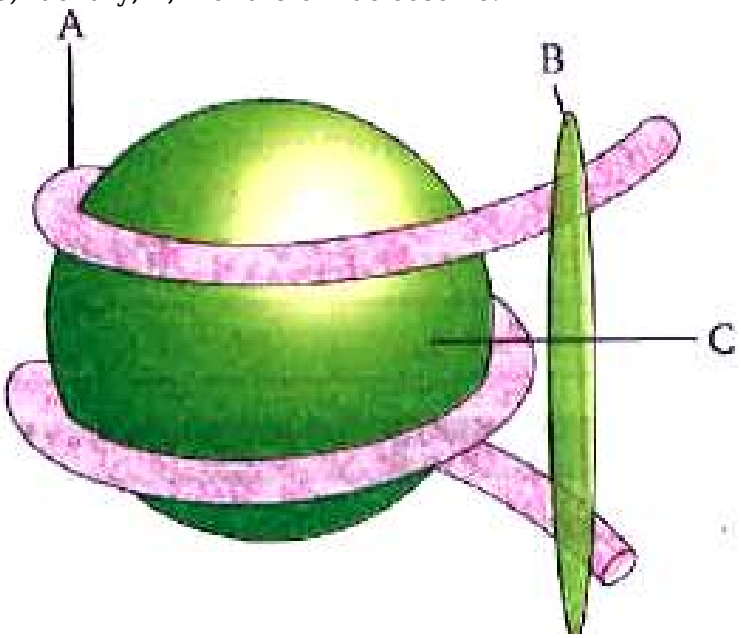
(1) PDB = 8, NGB = 6

(2) PDB = 3, NGB = 11

(3) PDB = 6, NGB = 8

(4) PDB = 8, NGB = 8

8) Identify, A, B and C of nucleosome:



(1) A = Histone core, B = DNA, C = H1

(2) A = DNA, B = Histone octamer, C = H1

(3) A = DNA, B = H1, C = Nu-body

(4) A = DNA, B = Histone core, C = H1

9) In an analysis of nucleotide composition of DNA, which of the following would be found?

(1) A = G and T = G

(2) A + C = G + T

(3) A = C

(4) G + C = T + A

10) While analysing the DNA of an organism a total number of 5386 nucleotides were found out of

which the proportion of different bases were: Adenine = 29%, Guanine = 1.7%, Cytosine = 32%, Thymine 17%. Considering the Chargaff's rule it can be concluded that:

(1) It is a double stranded circular DNA

(2) It is single stranded DNA

(3) It is a double stranded linear DNA

(4) No conclusion can be drawn

11) **Given below are two statements :**
Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.
Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.
In the light of the above statements, choose the correct answer from the options given below:

(1) **Statement I** is correct but **Statement II** is false.

(2) **Statement I** is incorrect but **Statement II** is true.

(3) Both **Statement I** and **Statement II** are true.

(4) Both **Statement I** and **Statement II** are false.

12) Read the following statements and choose correct statements :
A. Euchromatin is loosely packed chromatin.
B. Heterochromatin is transcriptionally active.
C. Histone octamer is wrapped by negatively charged DNA in nucleosome
D. Histones are rich in lysine and arginine
E. A typical nucleosome contains 400 bp of DNA helix
Choose the correct answer from the options below:

(1) (B); (D), (E) only

(2) (A), (C), (D) only

(3) (B), (E) only

(4) (A),(C),(E) only

13) **Assertion(A):** In a polynucleotide chain, nitrogenous bases are linked to the pentose sugar through N-glycosidic linkage.
Reason(R): The nitrogenous bases are attached to the 5' carbon Part of the pentose sugar in nucleotides.

(1) If both A and R are true and R is the correct explanation of A.

(2) If both A and R are true but R is not the correct explanation of A.

(3) If A is true but R is false.

(4) If A is false but R is true.

14) **Assertion(A):** The two strands of DNA have antiparallel polarity, with one strand having 5' to 3' polarity and the other 3' to 5'.
Reason(R): This antiparallel arrangement is necessary for the formation of hydrogen bonds between complementary bases.

(1) If both A and R are true and R is the correct explanation of A.

(2) If both A and R are true but R is not the correct explanation of A.

(3) If A is true but R is false

(4) If A is false but R is true.

15) In the transformation experiment, what did Griffith observe?

(1) Mixing heat-killed non-pathogenic strain of bacteria with living pathogenic strain makes the pathogenic strain non-pathogenic

(2) Mixing heat-killed pathogenic strain of bacteria with living non-pathogenic strain can convert all the living cells into the pathogenic form

(3) Mixing heat-killed pathogenic strain of bacteria with living non-pathogenic strain can convert some of the living cells into the pathogenic form

(4) Infecting mice with non-pathogenic strain of bacteria makes them resistant to pathogenic strains

16) The biochemical characterisation of 'transforming substance' in Griffith's experiment was done by:

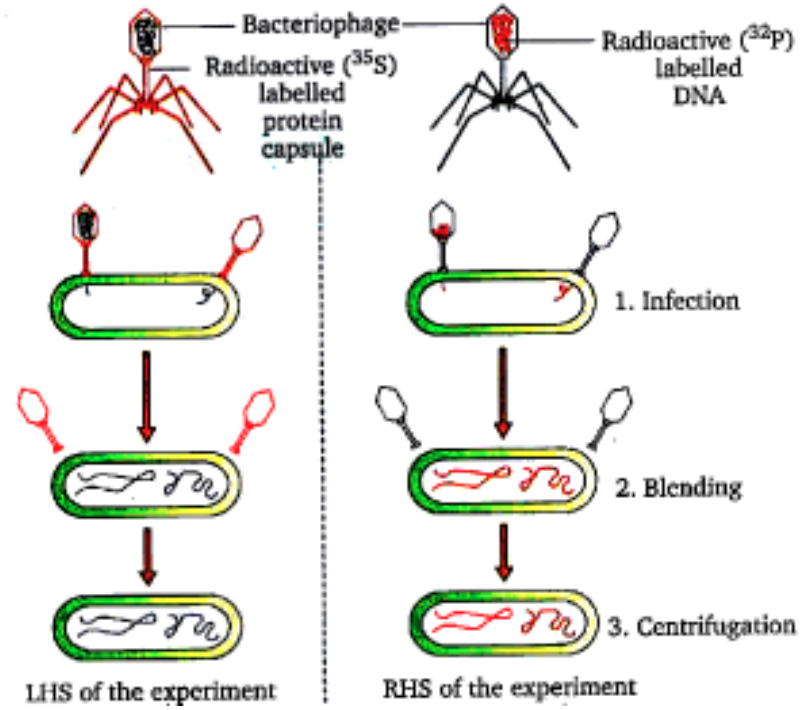
(1) Avery, Macleod and Mccarty, who proved the transforming substance is DNA

(2) Hershey and Chase, who proved the transforming substance is DNA

(3) Chargaff, who proved the transforming substance is DNA

(4) Avery, Macleod and Mccarty, who proved the transforming substance is protein

17) What is true regarding the left hand side(LHS) and right hand side(RHS) of the Hershey and Chase's experiment after centrifugation step is complete?



(1) LHS = Radioactivity detected in supernatant but not in cells
RHS = Radioactivity detected in supernatant but not in cells

(2) LHS = Radioactivity detected in cells but not in supernatant
RHS = Radioactivity detected in cells but not in supernatant

(3) LHS = Radioactivity detected in supernatant but not in cells
RHS = Radioactivity detected in cells but not in supernatant

(4) LHS = Radioactivity detected in cells but not in supernatant
RHS = Radioactivity detected in supernatant but not in cells

18)
Given below are two statements :
Statement I: RNA mutates at a faster rate.
Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is false.

(2) Statement I is incorrect but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

19) RNA is the genetic material in :

(1) TMV

(2) QB bacteriophage

(3) Corona virus and SARS virus

(4) All of the above

20) RNA was not selected as genetic material in most organisms because:

(1) It is chemically unstable

(2) The 2' OH group present at very nucleotide in RNA is a reactive group

(3) Some RNAs are catalytic

(4) All of the above

21) Read the following statements:
(i) RNA is chemically more stable than DNA
(ii) The presence of thymine at the place of uracil destabilizes DNA
(iii) RNA being stable mutates faster
(iv) DNA is preferred storage genetic material and for transmission of genetic information, RNA is better
(v) RNA was the first genetic material to evolve
Which of the above statements are correct?

(1) (i), (ii) and (iii)

(2) (iii) only

(3) (iv) only

(4) (iv) and (v)

22) Read the following statements :
(i) Some essential life processes like metaboism, translation and splicing evolved around RNA
(ii) During early evolution RNA used to act as genetic material as well as catalyst
(iii) DNA being double stranded and having complementary strand resist changes by evolving a

process of repair
(iv) DNA replication is semi-discontinuous and conservative
(v) Both DNA and RNA have the ability to direct their duplication
Which of the above statements are correct?

(1) (i), (iv) and (v)
(2) (iv) only
(3) (v) only
(4) (i), (ii), (iii) and (v)

23) **Assertion:** RNA was the first genetic material.
Reason: RNA was capable of acting both as genetic material and as a catalyst for biochemical reactions.

(1) If both A and R are true and R is the correct explanation of A.
(2) If both A and R are true but R is not the correct explanation of A.
(3) If A is true but R is false.
(4) If A is false but R is true.

24) Which one is incorrect statement?

(1) The basic unit of DNA is nucleotide.
(2) A nitrogenous base is linked to pentose sugar by N - glycosidic linkage
(3) Adenosine, guanosine and cytidines are nucleotides.
(4) DNA is the largest macromolecule of the cell.

25) In a DNA strand the nucleotides are linked together by

(1) Glycosidic bonds
(2) Phosphodiester bonds
(3) Peptide bonds
(4) Hydrogen bonds

26) Read the following statements and choose the correct option
I. Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are the two types of nucleic acids found in living systems.
II. DNA mostly functions as a messenger.
III. The genetic code determines the sequences of amino acids in proteins.
IV. DNA is the most abundant genetic material.
Correct statement(s) is/are

(1) Only I, II and III
(2) Only II, III and IV
(3) All the above
(4) Only I, III and IV

27) Read the following statements and choose the correct option.

Statement 1: A polymer of nucleotide has at one end a free phosphate moiety at 3' -end of sugar, which is referred to as 3'-end of polynucleotide chain.
Statement 2: At the other end of the polymer, the sugar has a free OH of 5C group which is referred to as 5' - end of the polynucleotide chain.

(1) Both statements are correct
(2) Only statement 1 is correct but statement 2 is incorrect
(3) Only statement 2 is correct but statement 1 is incorrect
(4) Both statements are incorrect

28) Central dogma of molecular biology was proposed by

(1) Francis Crick
(2) James Waston
(3) Ervin Chargaff
(4) Waston and Crick

29) If there are 48×10^8 base pairs in a eukaryotic cell, then what will be the number of beads-on-string structure present in the nucleus?

(1) 48×10^8
(2) 24×10^6
(3) 24×10^8
(4) 2.4×10^8

30) Nucleosomes consist of all except

(1) Histone octamer
(2) Lysine -and arginine rich proteins
(3) Polyamines
(4) 200 base pairs

31) Select the option that correctly fill up the blanks in the following statements.
(i) In RNA, the uracil is found at the place of_____A_____.
(ii) The pitch of the DNA double helix is ___B_____
(iii) In a DNA double helix, the backbone is constituted by sugar phosphate and the bases project _____C_____
(iv) The length of DNA in diploid human cell is____D____

(1) A-Thymine; B-34 nm; C-Outside; D-1.2 m
(2) A-Thiamine; B-3.4 nm; C-Outside; D-2.2 m
(3) A-Thiamine; B-0.34 nm; C-Inside; D-1.2 m
(4) A-Thymine; B-3.4 nm; C-Inside; D-2.2 m

32) The fact that a purine base always paired through hydrogen bonds with a pyrimidine base in the DNA double helix, leads to

(1) The antiparallel nature

(2) The semi-conservative nature

(3) Uniform Length throughout DNA

(4) Uniform length in all DNA

33) Read the following statements with respect to the structure of DNA, and state true (T) or false (F).
(A) It is made up of two polynucleotide chains that have antiparallel polarity.
(B) The two chains are coiled in a right-handed fashion.
(C) Phosphate group is linked to 5'-OH of the same nucleoside by glycoside linkage for the formation of sugar phosphate backbone.
(D) All four deoxyribonucleotides are always equally present in both the strands

(1) A - T; B - T; C - F; D - F

(2) A - T; B - T; C - T; D - F

(3) A - F; B - F; C - T; D - T

(4) A - T; B - F; C - F; D - T

34) The beads-on-strings is/are

(1) Exons interrupted by introns

(2) Nucleosomes in chromatin

(3) Smaller submit of ribosomes in polysome

(4) Amino acid in a long polypeptide chain

35) Who amongst the following scientists has no contribution in the development of the double helix model for the structure of DNA?

(1) Rosalind Franklin

(2) Maurice Wilkins

(3) Erwin Chargaff

(4) Meselson and Stahl

36) In the polynucleotide chain of DNA, a nitrogenous base is linked to the OH of

(1) 1' C sugar

(2) 2' C sugar

(3) 3' C sugar

(4) 4' C sugarNo conclusion can be drawn.

37) A molecule that can act as a genetic material must fulfill the following criteria given below except

(1) It should be able to generate its replica(Replication)

(2) It should be chemically and structurally stable

(3) It should provide the scope for slow changes(mutation) that are required for evolution

(4) It should not be able to express itself in the form of 'mendelian characters'

38) Griffith could distinguish the two strains of *Streptococcus pneumoniae* due to

(1) Colony appearance in culture
(2) Difference in their lethality in mice
(3) Their size
(4) All except(3)

39) Mark the odd one out with respect to chargaff's rule

(1) A + G = T + C
(2) $\frac{A+T}{G+C}$ ration is more than one in E.coli
(3) It is applicable to ds DNA only
(4) Sugar deoxyribose & phosphates occur in equimolar portion

40) The unequivocal proof that DNA is the genetic material came from the experiment of _____

(1) Miescher
(2) Alfred Hershey and Martha Chase
(3) Frederick Griffith
(4) James Waston and Francis Crick

41) Consider the following statements regarding transformation experiment.
(A) It was performed by F. Griffith on *Streptococcus pneumoniae*.
(B) It proved that DNA is the genetic material.
(C) Avery, Macleod, and McCarty worked to determine the biochemical nature of transforming principle.
(D) DNase inhibits the transformation.
The correct statement is/are

(1) (A) and (B) only
(2) (A) only
(3) (A), (B), and (C) only
(4) (A), (B), (C), and (D)

42) Read the following statements about Hershey and Chase experiment and choose the correct option.
I. Viruses grown in the presence of radioactive phosphorus contained radioactive protein.
II. Viruses grown on radioactive sulphur contained radioactive DNA.
III. Bacteria which was infected with viruses that had radioactive DNA were radioactive, indicating that DNA was the material that passed from the virus to the bacteria.
IV. Bacteria that were infected with viruses that had radioactive proteins were not radioactive. This indicates that proteins did not enter the bacteria from the viruses.
Correct statement(s) is/are

(1) All the above

(2) Only III and IV

(3) Only I and II

(4) Only I and III

43) Teminism is concerned with _____ activity.

(1) Reverse transcriptase

(2) DNA -dependent RNA polymerase

(3) RNA -dependent DNA polymerase

(4) Both (1) and (3)

44) Match the columns and select the correct option.

	Column I (Scientist)		**Column II (Experimental material)**
a.	F. Griffith	**1.**	*Vicia faba*
b.	Meselson and stahl	**2.**	Bacteriophages
c.	Taylor	**3.**	*Streptococcus pneumoniae*
d.	Hershey and chase	**4.**	*Escherichia coli*

(1) a - 1, b - 2, c - 3,.d - 4

(2) a - 4, b - 3, c - 2,.d - 1

(3) a - 3, b - 4, c - 1,.d - 2

(4) a - 2, b - 1, c - 4,.d - 3

45) Match the column and select the correct option.

	Column I		**Column II**
a.	Double - helical model of DNA	**1.**	Francis Crick
b.	Transforming principle	**2.**	Jacob and Monod
c.	Central dogma	**3.**	Frederick Griffith
d.	Lac operon model	**4.**	Waston and Crick

(1) a - 1, b - 3, c - 4, d - 2

(2) a - 4, b - 3, c - 1, d - 2

(3) a - 2, b - 4, c - 3, d - 1

(4) a - 3, b - 1, c - 2, d - 4

46) How many of following disease is associated with cigarette smoking ?
Lung cancer, Throat cancer, Bronchitis, Emphysema, Gastric ulcer, amnesia, Psychosis

(1) Four

(2) Five

(3) Six

(4) Seven

47) How many of the following statements are incorrect about tobacco addiction and its effects?
(a) It can be chewed or smoked but not snuffed.
(b) Tobacco contains alkaloids.
(c) Tobacco chewing is associated with increased risk of cancer of oral cavity.
(d) It increases carbon monoxide content as well as concentration of haem-bond oxygen in blood.

(1) One

(2) Three

(3) Two

(4) Four

48) Which of the following statements regarding drugs are correct?
(i) Heroin, commonly called smack, is obtained by acetylation of morphine.
(ii) Cocaine is obtained from the latex of Papaver somniferum.
(iii) Marijuana interferes with the transmission of dopamine.
(iv) Morphine is an effective painkiller.

(1) (i) and (ii)

(2) (i) and (iv)

(3) (ii) and (iii)

(4) (iii) and (iv)

49)

Which of the following combination is not a cannabinoids?

(1) Charas, Hashish

(2) Marijuana, Bhang

(3) Charas, Ganja

(4) Datura, LSD

50) Which of the following is incorrect match regarding the drug and its sources?

(1)	LSD	Fruiting body of fungus *Cleviceps Purpurea*
(2)	Opium	Latex unripe capsule of *Papaver Somniferum*
(3)	Hashish	Dried Leaves of *Erythroxylum Coca*
(4)	Ganja	Dried unfertilised female inflorescence of Cannabis plant

(1) 1

(2) 2

(3) 3

(4) 4

51) Column-A has some drugs and column-B has their description. Select the option which has

correct match :

Coumn-A		Column-B	
P	Morphine	V	Abused by sports person
Q	Cannabinoid	W	Used in patients of mental illness
R	Cocaine	X	Produce a sense of euphoria
S	Barbiturates	Y	Stimulate adrenal gland
T	Nicotine	Z	Sedative and pain killer

(1) P-V Q-Z R-W S-X T-Y

(2) P-Z Q-V R-X S-W T-Y

(3) P-Z Q-W R-X S-V T-Y

(4) P-Y Q-X R-V S-W T-Z

52) Choose the incorrect matching pair.

(1)	Cannabinoids	effect on cardiovascular system
(2)	Coca alkaloid	potent stimulating action on CNS
(3)	Coke and crack	slow down body functions
(4)	Cocaine	sense of euphoria

(1) (1)

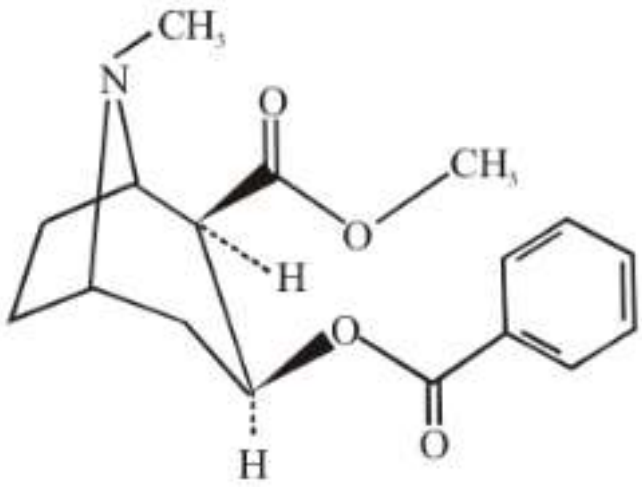
(2) (2)

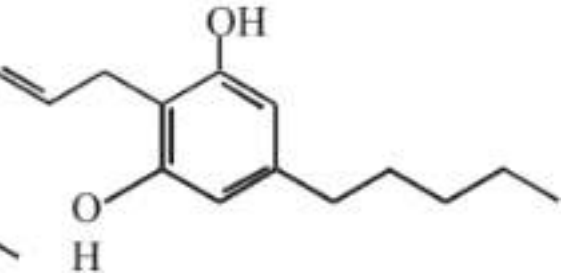
(3) (3)

(4) (4)

53) Identify the molecules (a) and (b) given below and select the right option giving their source and use.

(a)



(b) 

	Molecule	Source	Use
(1)	(a) Cocaine	*Erythroxylum coca*	Accelerates the transport of dopamine
(2)	(b) Diacetyl morphine	*Cannabis sativa*	Depressant and slows down body functions
(3)	(b) Cannabinoid	*Atropa belladona*	Produces hallucinations
(4)	(a) Morphine	*Papaver somniferum*	Pain killer

(1) (1)

(2) (2)

(3) (3)

(4) (4)

54) Find the correct statement about tobacco smoking –
(A) Nicotine inhibits secretion of adrenal medulla.
(B) Smoking is associated with increased incidence of cancer of lungs and throat.
(C) Smoking increases carbon monoxide content in blood and decreases the concentration of haembound oxygen.
(D) Smoking decreases blood pressure.

(1) A and D

(2) B and D

(3) A and B

(4) B and C

55) (a) Smack
(b) Diacetylmorphine
(c) White
(d) Odourless
(e) Bitter crystalline compound
Above statements/informations are correct for:-

(1) Morphine

(2) Heroin

(3) Cocain

(4) Barbiturates

56) Mark the incorrect statement :

(1) Smoking also paves the way to hard drugs.

(2) Smoking is associated with liver cirrhosis.

(3) Smoking increases carbon monoxide (CO) content in blood and reduces the concentration of haembound oxygen

(4) Nicotine stimulates adrenal gland to release adrenaline and nor-adrenaline into blood circulation, both of which raise blood pressure and increase heart rate.



57) Choose correct option about given below :-

	A	B	C
(1)	Hallucinogen	Bhang	Heroin
(2)	Morphine	Datura	Hallucinogen
(3)	Heroin	Bhang	Hallucinogen
(4)	Bhang	Hallucinogen	Morphine

(1) 1

(2) 2

(3) 3

(4) 4

58) Use of following compounds in females produces side effects such as masculinisation, excessive hair growth on face and body :

(1) Narcotic analgesics

(2) Anabolic steroid

(3) Diuretic drugs

(4) All of the above

59) Mark the correctly matched options :-

(A)	Leaves of	–	 Cannabis sativa
(B)	Opium poppy	–	

(C)	Flowing branch	–	 of Datura

(1) A and B

(2) B and C

(3) A, B and C

(4) Only C

60) Read the following statements (A-D) :
(A) Addiction is a psychological attachment to certain effects associated with drugs and alcohol.
(B) With repeated use of drugs, the tolerance level of the receptors present in our body increases.
(C) Use of drugs even once, can be a fore-runner to addiction.
(D) Dependence is the tendency of the body to manifest a characteristic and unpleasant withdrawal syndrome if regular does of drugs/alcohol is abruptly discontinued.
How many of the above statements are **true**

(1) 4

(2) 3

(3) 2

(4) 1

61) Given below are two statements:
Statement I: The addictive potential of drugs and alcohol can lead to their regular use and abuse, pulling the user into a vicious circle.
Statement II: Dependence is characterized by a pleasant withdrawal syndrome when the regular dose of drugs/alcohol is abruptly discontinued.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true.

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false.

(4) Statement I is true but Statement II is false.

62) Withdrawal syndrome is characterised by :-
(A) Anxiety (B) Shakiness
(C) Nausea (D) Sweating

(1) A, C

(2) B, C

(3) A, C, D

(4) A, B, C, D

63) Adolescence period is between the age of

(1) 10 - 15 yrs

(2) 12 - 18 yrs

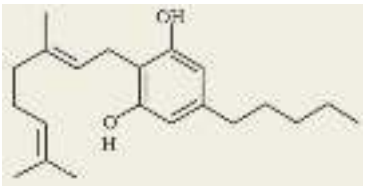
(3) 18 - 23 yrs

(4) 8 - 13 yrs

64) Which pair is matched correctly?

(1)  ⇒ Flowering branch of *Datura*

(2)  ⇒ Leaves of *cannabis sativa*

(3) 

⇒ Smack

(4)  ⇒ Opium poppy

65) Observe the following table very carefully contain source of drug, biological name and their uses. Choose the correct option :

(1)		*Papaver somniferum*	Stimulant action on body
(2)		*Claviceps purpura*	Hallacination

(3)		*Papaver somniferum*	Depressant action on body
(4)		*Cannabis sativa*	Relieving pain

(1) 1

(2) 2

(3) 3

(4) 4

66) Odd one out regarding side effects of using anabolic steroids in female sportspersons

(1) Breast enlargement

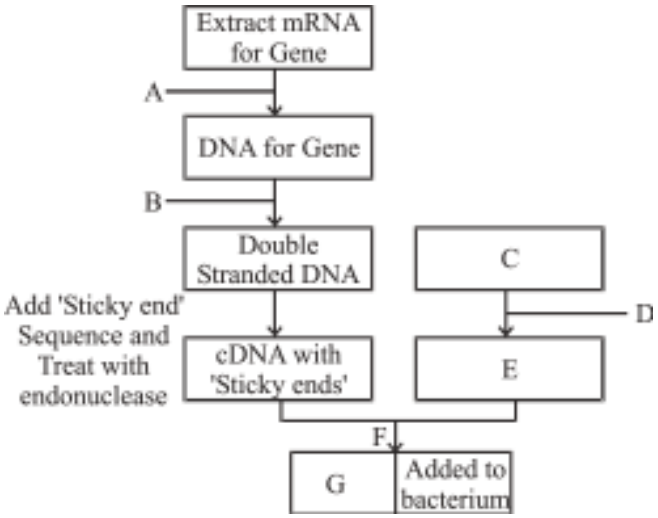
(2) Excessive hair growth on face and body

(3) Deeping of voice

(4) Masculinisation

67)

Identify the labelled items A, B, C, D, E, F and G in the diagram below from the list I to VII :-



(I) DNA polymerase

(II) Plasmid

(III) Plasmid with 'sticky ends'

(IV) DNA ligase

(V) Restriction endonuclease

(VI) Recombinant DNA

(VII) Reverse transcriptase The correct components are

	A	B	C	D	E	F	G
(1)	VII	I	II	V	III	IV	VI
(2)	VII	VI	V	IV	III	II	I

(3)	VII	V	III	I	II	IV	VI
(4)	I	II	IV	VI	III	V	VII

(1) 1

(2) 2

(3) 3

(4) 4

68) Superbug is

(1) Widely used technique in gene therapy and cancer treatment

(2) Genetically engineered aquatic bug having oil digesting enzyme forming genes

(3) Genetically engineered bacterium for production of Bt-corn

(4) Widely used technique to increase oil production

69) Arrange the steps of rDNA technology in correct order:-
I. Extraction of the desired gene product.
II. Amplification of gene of interest.
III. Isolation of desired DNA fragment.
IV. Ligation of DNA fragment into vector.
V. Insertion of rDNA into host.
Correct order is :-

(1) I, II, III, IV, V

(2) V, IV, III, II, I

(3) III, II, IV, V, I

(4) III, IV, II, I, V

70) Given below are two columns. In Column I is the list of four enzymes and in Column II is the list of functions of the given enzymes. Which one of the following options shows the enzymes matched with their respective functions correctly?

	Column I (Enzyme)		**Column II (Function)**
P.	DNA Ligase	**i.**	Removes nucleotides from ends of DNA
Q.	Restriction exonuclease	**ii.**	Extends primer on a DNA template
R.	Taq polymerase	**iii.**	Joins the DNA fragments
S.	Restriction endonuclease	**iv.**	Cuts DNA at a specific position

Options :

(1) P-i, Q-ii, R-iv, S-iii

(2) P-iv, Q-iii, R-ii, S-i

(3) P-i, Q-iv, R-iii, S-ii

(4) P-iii, Q-i, R-ii, S-iv

71) Refer to the following statements for isolation of genetic material :-

(a) In majority of organisms deoxyribonucleic acid or DNA is the genetic material.
(b) In order to cut the DNA with restriction enzymes, it needs to be in pure form, free from other macro-molecules.
(c) The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease.
(d) Purified DNA ultimately precipitates out after the addition of chilled methanol.
Choose the most appropriate answer from the options given below :-

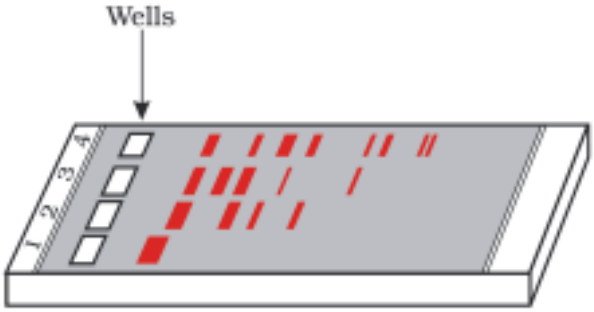
(1) (a) and (b) only

(2) (a), (b) and (c) only

(3) (a), (b) and (d) only

(4) (b), (c) and (d) only

72) Match the following columns according to the below diagram :-



Column-I		Column-II	
(a)	Well-1	(i)	Closest to cathode
(b)	Well-2	(ii)	Medium restriction site
(c)	Well-4	(iii)	Undigested DNA
(d)	Smallest DNA	(iv)	Closest to anode
(e)	Largest DNA	(v)	Maximum restriction site

(1) (a)-(iii), (b)-(ii), (c)-(v), (d)-(iv), e-(i)

(2) (a)-(iii), (b)-(ii), (c)-(v), (d)-(i), e-(iv)

(3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(v), e-(iv)

(4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(v), e-(i)

73) To ligate an Alien DNA, atleast how many antibiotic resistance should be present on plasmid?

(1) One

(2) Two

(3) Three

(4) Four

74) **Statement I :** EcoRI cuts the DNA between bases G and A only when the sequence GAATTC is present in the DNA
Statement II : EcoRI cut the strand of DNA a little away from the centre of the palindrome sites, but between the same two bases on the opposite strands.

(1) Statement I is correct & Statement II is incorrect

(2) Statement I is incorrect & Statement II is correct

(3) Both Statements are correct

(4) Both Statements are incorrect

75) If a recombinant DNA bearing gene for ampicillin resistance is transferred into *E.coli* cells and the host cells are spread on agar plates containing ampicillin, then :-

(1) Both transformed and untransformed recipient cells will die

(2) Both transformed and untransformed recipient cells will grow

(3) Transformed recipient cells will grow and untransformed recipient cells will die

(4) Transformed recipient cells will die and untransformed recipient cell will grow

76) Fill up the blanks :
A. When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of ..... and these can be joined together end-to-end using .....
B. Disruption of the cell membranes can be achieved by treating the bacterial cells, plant or animal tissue with enzymes, such as ..... (for bacteria), ... (for plant cells) and ..... (for fungus).
C. If any protein encoding gene is expressed in a heterologous host, it is called a ..... .
D. In gel electrophoresis, the separated DNA fragments can be seen only after staining the DNA with a compound known as ..... followed by exposure to UV radiation.

(1) A - sticky ends, Recombinase; B - Lysozyme, Cellulase, Chitinase; C - Recombinant protein; D - Ethidium bromide

(2) A - sticky ends, DNA ligases; B - Exonuclease, Cellulase, Chitinase; C - Recombinant protein; D - Ethidium bromide

(3) A - sticky ends, DNA ligases; B - Lysozyme, Cellulase, Chitinase; C - native protein; D - Cesium chloride

(4) A - sticky ends, DNA ligases; B - Lysozyme, Cellulase, Chitinase; C - Recombinant protein; D - Ethidium bromide

77) Among many, the two core techniques that enabled birth of modern biotechnology are :

(1) Genetic engineering and Human Genome project

(2) Genetic engineering and Bioprocess engineering

(3) Chemical engineering and ELISA

(4) Gene cloning and gene transfer

78) **Assertion :** A plasmid can be used as vector to deliver an alien piece of DNA into the host organism.
Reason : Plasmids are autonomously replicating circular extra chromosomal DNA.

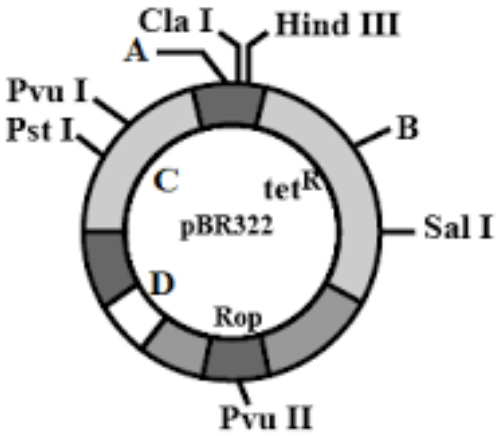
(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

79) Identify A, B, C and D in the given figure of *E. coli* cloning vector pBR322 and select the correct option :



(1) A - Hind II, B - EcoR I, C - Amp^R, D - ori

(2) A - Hind I, B - BamH I, C - Ori, D - EcoRI

(3) A - EcoR I, B - Ori, C - Pst II, D - Amp^R

(4) A - EcoR I, B - BamH I, C - Amp^R, D - ori

80) If in a linear DNA of 12 kb size, two recognition sites of *Bam*HI are present 4 kb distance apart and one of them is at 4 kb distance from one end. Then, how many bands will be obtained on separation of DNA fragments using agarose gel electrophoresis, assuming complete digestion by *Bam*HI?

(1) Three

(2) Two

(3) One

(4) Four

81) Match the columns :-

	Column-A		**Column-B**
(1)	Elution	(A)	Cloning vector
(2)	Ampicillin resistant gene	(B)	Obtaining DNA Fragments with sticky end
(3)	Ti Plasmid	(C)	Selectable marker
(4)	Restriction enzyme	(D)	Obtaining bands of DNA from agrose gel

(1) 1-C, 2-D, 3-A, 4-B

(2) 1-D, 2-C, 3-B, 4-A

(3) 1-D, 2-C, 3-A, 4-B

(4) 1-C, 2-D, 3-B, 4-A

82) Read the following statements carefully.
(a) The first restriction endonuclease - BamHI
(b) Restriction enzymes are used in the isolation of DNA from other macro-molecule.

(c) Downstream processing is one of the steps of r-DNA technology.
(d) Disarmed pathogen vectors are also used in the transfer of r-DNA into the host.
How many statements is/are **correct**.

(1) Two
(2) One
(3) Three
(4) Four

83) **Statement-I :-** The construction of first r-DNA emerged from the possibility of linking a gene encoding antibiotics resistance with a native plasmid of *Agrobacterium tumifaciences*.
Statement-II :- Stanley Cohen and Herbert Boyer accomplished r-DNA in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for production of vitamine A.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and Statement-II are incorrect
(3) Statement-I is incorrect and statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

84) The integration of how many of the following for products and services is called biotechnology?
natural science, organisms, cells, parts of cells, molecular analogues

(1) Five
(2) Four
(3) Three
(4) Two

85) The colonies of recombinant bacteria appear white in contrast to blue colonies of nonrecombinant bacteria because of -

(1) Insertional inactivation of β-galactosidase in recombinant bacteria
(2) Inactivation of glycosidase enzyme in recombinant bacteria
(3) Non-recombinant bacteria containing beta galactosidase
(4) Insertional inactivation of β-galactosidase in non-recombinant bacteria

86) If we ligate foreign DNA at Bam HI site of pBR322, then the resultant recombinants will show
(a) Resistance to ampicillin
(b) Sensitivity to tetracycline
(c) Resistance to tetracycline
(d) Sensitivity to ampicillin

(1) a and b
(2) a and c
(3) b and d
(4) c and d

87) In pBR322 amp^R site have recognition sequence for -

(1) Pvu I and BamH I

(2) Pvu I and Pst I

(3) BamH I and Sal I

(4) BamH II and Hind III

88) Read the following sentences.
(a) In elution the separated bands of DNA are cut out from agarose gel and extracted from the gel piece
(b) *E.coli* cloning vector pBR-322 has several restriction sites, an *ori,* antibiotic resistance genes and *rop*.
(c) The downstream processing and quality control testing vary from product to product.
(d) Competent bacterial cells take up the plasmid.

(1) All are incorrect

(2) Except d, all are correct

(3) Only d is correct

(4) All are correct

89) Select the correct match :

(1)	Insertional inactivation	-	Differentiate recombinants from non recombinants
(2)	Recombinant protein	-	Any protein encoding gene is expressed in a homologous host.
(3)	Down stream processing	-	A series of process which start from raw material and to completion of biosynthetic product
(4)	Cloning	-	Introduction of the identified DNA into the host.

(1) 1

(2) 2

(3) 3

(4) 4

90) Match the column -

Column-I		**Column-II**	
(a)	Gel electrophoresis	(i)	Bright orange coloured
(b)	PCR	(ii)	Chilled ethanol
(c)	DNA precipitation	(iii)	Sieving effect

(d)	Ethidium bromide stained DNA	(iv)	DNA primer

(1) a - i; b - ii, c - iii, d - iv

(2) a - ii; b - iii, c - iv, d - i

(3) a - iii; b - iv, c - i, d - ii

(4) a - iii; b - iv, c - ii, d - i

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	3	1	4	4	3	2	1	3	2	3	3	4	2	1	1	2	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	4	1	3	4	1	2	1	1	3	1	2	3	1	3	1	2	2	4
Q.	41	42	43	44	45															
A.	1	1	3	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	2	2	2	1	1	2	1	3	1	3	2	4	1	1	3	1	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	3	2	3	2	4	2	4	2	2	1	3	3	4	4	3	2	1	2
Q.	86	87	88	89	90															
A.	3	3	2	1	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	4	1	4	1	1	3	3	2	2	2	2	3	1	3	1	3	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	2	3	2	4	4	1	2	3	4	3	1	2	4	1	4	4	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	4	2	2	2	3	2	4	3	2	(3)	(4)	4	2	2	3	2	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	2	2	3	1	1	2	3	4	2	1	2	3	3	4	2	1	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	2	1	1	1	2	4	1	4										

SOLUTIONS

PHYSICS

1) $H = \frac{1}{2}Li_0^2 = \frac{1}{2}L\left(\frac{E}{R_1}\right)^2$

2) $\varepsilon_L = L\frac{di}{dt}$

3) $e = -\frac{d\phi}{dt}$

$e = -a\left(n^2-1\right)t^{n^2-2}$

$\Rightarrow e = a\left(n^2-1\right)t^{n^2-2}$

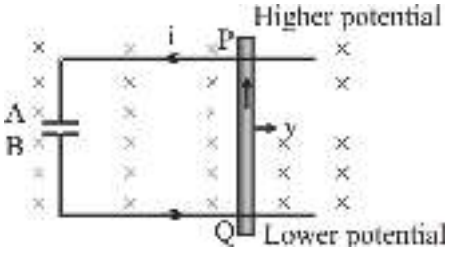
For n = ± 1, e = 0

For $n^2 - 1 = 1$ we get ϕ = at i.e. |e| = a ⇒ $n = \pm\sqrt{2}$ for |e| = a

4)

$Q = CV = C(Bvl) = 10 \times 10^{-6} \times 4 \times 2 \times 1 = 80\ \mu C$

According to Fleming's right hand rule induced current flows from Q to P. Hence P is at higher potential and Q is at lower potential. Therefore A is positively charged and B is negatively charged.



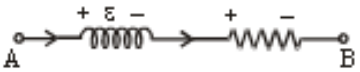
5) $E_P = 240$ V $E_S = 24$ V, $E_S I_S = 24$ W

Current in primary coil $I_P = \frac{E_S I_S}{E_P} = \frac{24}{240} = 0.1A$

6) ϕ = Mi

BAcos0° = Mi

$M = \frac{2\sqrt{2}\mu_0 l^2}{\pi L}$

7) 

$V_A - L\frac{di}{dt} - iR = V_B$

$V_A - V_B = L\frac{di}{dt} + iR$

8) $\varepsilon = -\frac{d\phi}{dt}$

9)

Initial slope = $\frac{I_0}{\tau}$ or $\frac{V}{L}$
as from A to B, initial slope decreases, hence L increase with E & R constant.

10) $i = i_0[1-e^{-\frac{t}{\lambda}}]$

$i = \frac{i_0}{4}$

$\Rightarrow \frac{i_0}{4} = i_0\left[1 - e^{-\frac{t}{\lambda}}\right]$

$\Rightarrow \frac{1}{4} = 1 - e^{-\frac{4}{\lambda}}$

$\Rightarrow e^{-\frac{4}{\lambda}} = \frac{3}{4}$

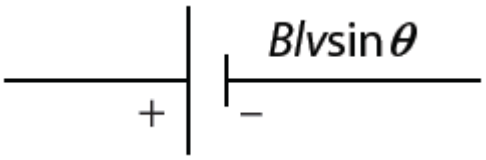
$\Rightarrow e^{4/\lambda} = \frac{4}{3}$

$\Rightarrow \frac{4}{\lambda} = \log_e\left(\frac{4}{3}\right)$

$\Rightarrow \lambda = \frac{4}{\log_e\left(\frac{4}{3}\right)}$

11) $e = M\frac{dI}{dt}$
straight line, I = Kt
e = constant

12)



Sol:

13)

Sol: 45 A

14)

Sol: $BR^2\omega/2$

15)

Force an loop in these condition.

$F = \frac{B^2\ell^2 v}{R}$

$\Rightarrow V = \frac{FR}{B^2\ell^2}$

16) $V_A - 5 + 15 + 5 \times 10^{-3} \times 10^3 = V_B$

17)

$e_0 = mI_0w$ (max)
$5\pi = m \times 10 \times 100\pi$
$m = 5$ mH

18) $I = I^0 (1 - e^{-t/l})$ $\left(\lambda = \frac{L}{R} = \frac{10}{5} = 2 \text{ sec}\right)$

$= \frac{E}{R}(1 - e^{-t/l}) = 1(1 - e^{-2/2}) = (1 - e^{-1})$

19)

Sol: $\frac{B^2AL}{2\mu_0}$



20)
$e = B(v\sin\theta)(2R)$

21)

Twice per revolution

22) $\vec{A} = 25cm^2 = 25 \times 10^{-4} m^2 \hat{k}$
$\vec{B} = 0.6\left(-\hat{k}\right)$
$\varepsilon = -\frac{d\phi}{dt} = \frac{\Delta\phi}{\Delta t}$

23)

$$0.6 = \frac{R_1}{\sqrt{R_1^2 + X_L^2}}$$

$$R_1^2 + X_L^2 = \frac{25}{9}R_1^2$$

$$X_L = \frac{4}{3}R_1 \quad 0.5 = \frac{R_2}{\sqrt{R_2^2 + X_C^2}}$$

$$X_C = R_2\sqrt{3}$$

$\because$ $X_C = X_L$ as $\cos\phi = 1$

$$R_2\sqrt{3} = \frac{4}{3}R_1 \Rightarrow \frac{R_1}{R_2} = \frac{3\sqrt{3}}{4}$$

24) **Explain question:** Question asking for RMS value of voltage.

Concept: This question is based on alternating current RMS value.

Solution: $V_{rms} = \sqrt{\bar{V}^2} \Rightarrow V_{rms}^2 = \bar{V}^2 = \frac{\int_{t_1}^{t_2} V^2 dt}{t_2 - t_1}$

$\therefore$ $V_{rms}^2 = \frac{\int_0^1 \left(V_0 t^{3/2}\right)^2 dt}{1-0} = V_0^2 \int_0^1 t^3 dt$

therefore, $V_{rms} = \frac{V_0}{2}$

Final Answer: Option 1 is correct.

25) $X_L = L\omega$, $\omega \uparrow$ $X_L \downarrow$

$X_C = \frac{1}{\omega C}$, $\omega \uparrow$ $X_C \downarrow$

26)

Sol: 100V, 4A

27)

Sol: $\sqrt{i_0^2 + 0.5i_1^2}$

28) $P = \frac{V^2}{z^2}R$

$z = \sqrt{R^2 + (L\omega)^2}$

$$P = \frac{V^2 R}{\left(R^2 + L^2\omega^2\right)}$$

29) Conceptual

30) $V = L\frac{dI}{dt}$

$\Rightarrow \frac{dI}{dt} = \frac{V_0}{L}\sin\omega t$

$\Rightarrow I = -\frac{V_0}{\omega L}\cos\omega t$

Instantaneous power,

$P = VI = --\frac{{V_0}^2}{\omega L}\sin\omega t\cos\omega t$

$= -\frac{{V_0}^2}{2\omega L}\sin 2\omega t$

31)

Sol: 90%

32) With decrease in frequency ω inductive reactance $X_L = \omega L$ decrease and capacitive reactance $X_C = \frac{1}{\omega C}$ increase.

33)

Sol: $\frac{\sqrt{3}}{\pi}H$

34)

Sol: $P\left(\frac{R}{Z}\right)^2$

35)

Sol: 60 V, 30 V, 30 V

36)

Sol: 100 $\sqrt{6}$ watt

37)

Conceptual

$I = I_0 \sin \omega t$

$I_{rms} = \frac{I_0}{\sqrt{2}}$

So $\frac{I_0}{\sqrt{2}} = I_0 \sin \omega t$

$\sin \frac{\pi}{4} = \sin(2\pi)(50)t$

$\frac{\pi}{4} = 2\pi(50)t$

$t = \frac{1}{400}$ s

38) $t = 0.25 \times 10^{-2} \Rightarrow 2.5 \times 10^{-3}$ s

39)

Conceptual

40)

Given resonance condition and $V_L = V_C = V_R = 10$ volt
So $X_L = X_C = R$
If current is I

$$I = \frac{V}{Z} = \frac{V}{R}$$

If resistance is halved

$$I' = \frac{V}{\frac{R}{2}} = \frac{2V}{R} = 2I$$

$V_L = 2I\ X_L$, $V_C = 2IX_C$
Hence $V_L = V_C = 20$ volt

$V_R = (2I)\left(\frac{R}{2}\right) = IR = 10$ volt

41)

$2\pi f = 75\pi$
$2f = 75$
current becomes zero two times in one cycle.

42)

for frequency f

$Z_1 = \sqrt{R^2 + X^2}$

For frequency 2f,

$X' = \frac{x}{2}$

as for R – C, $X = \frac{1}{\omega c} = \frac{1}{2\pi fc}$

$Z_2 = \sqrt{R^2 + \left(\frac{X}{2}\right)^2}$

Then $\frac{Z_1}{Z_2} = \frac{\sqrt{R^2 + X^2}}{\sqrt{R^2 + \frac{X^2}{4}}}$

For X << R, $\frac{Z_1}{Z_2} = 1$

For X >> R, $\frac{Z_1}{Z_2} = 2$

So in general $\frac{Z_1}{Z_2}$ is between 1 and 2

43) p.f. = $\cos\phi = \frac{1}{\sqrt{2}} = \frac{R}{Z}$

$\Rightarrow Z = \sqrt{2}R \Rightarrow Z = 10\sqrt{2}\Omega$

Here $Z = \sqrt{R^2 + (X_C - X_L)^2}$

where X_L = = ωL = (100)(0.1) = 10 Ω

$200 = 100 + (X_C - 10)^2$

$\Rightarrow X_C = 20\Omega$

$\Rightarrow \frac{1}{100C} = 20$

$\Rightarrow C = 500\ \mu F$

44)

Conceptual

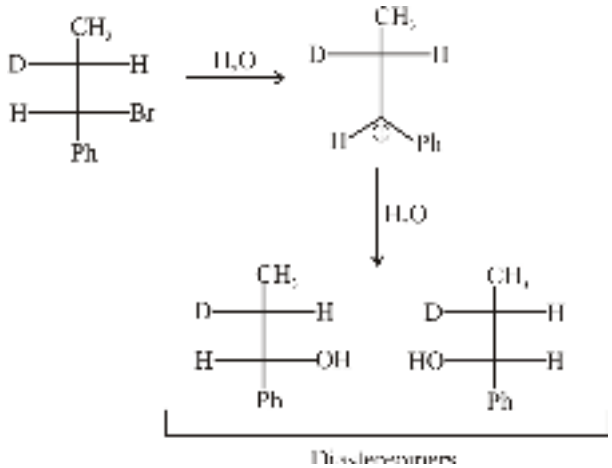
45) $P = \frac{v_0^2 R}{2\left(\omega^2 L^2 + R^2\right)}$

$\frac{dP}{dR} = 0 = \frac{2\left(\omega^2 L^2 + R^2\right) v_0^2 - 2v_0^2 R \times 2R}{4\left(\omega^2 L^2 + R^2\right)^2} = 0$

$R = \omega L = \frac{5}{100\pi} \times 100\pi = 5\Omega$

CHEMISTRY

46)



47)

Conceptual

48)

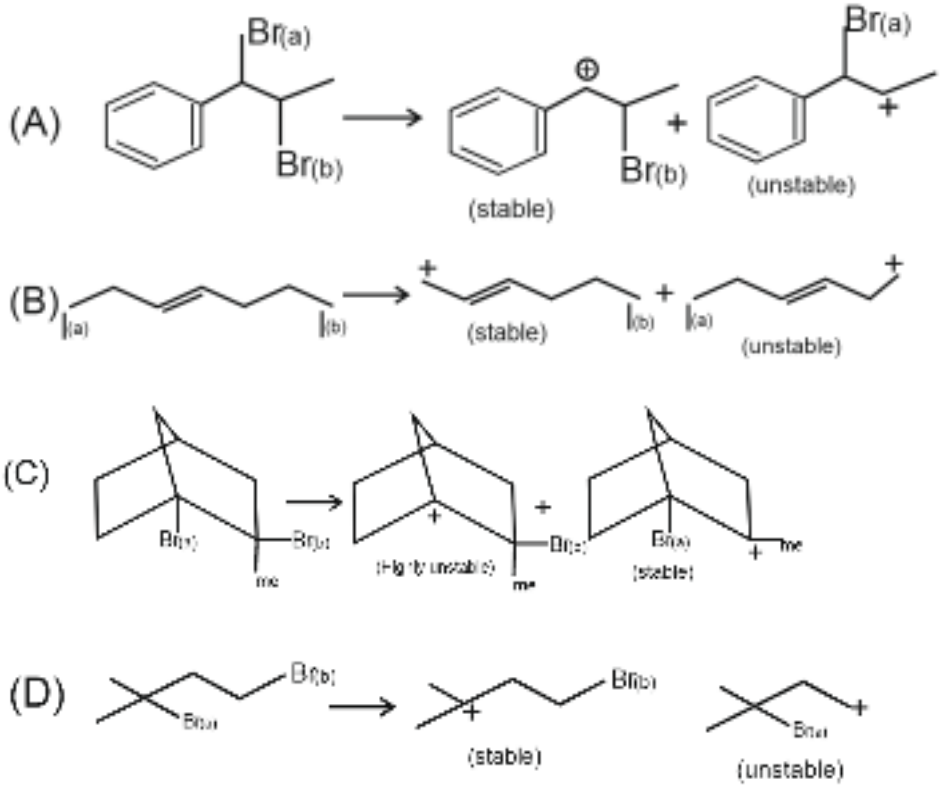
Conceptual

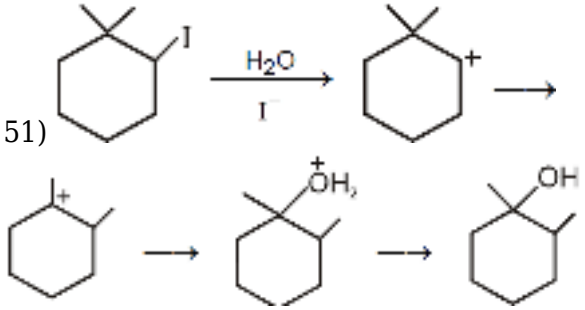
49)

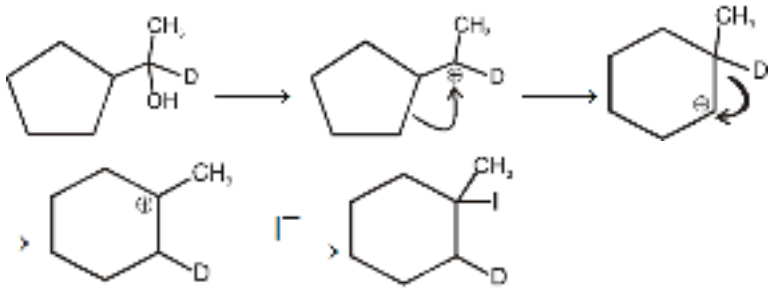
Conceptual

50)

Reactivity of SN^1 reaction α stability of carbocation

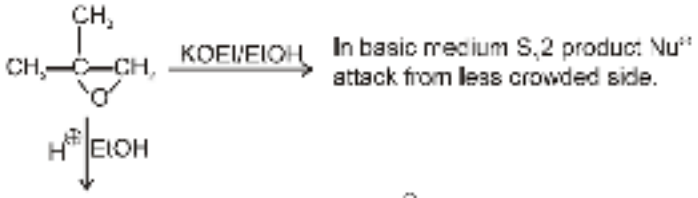


51) 

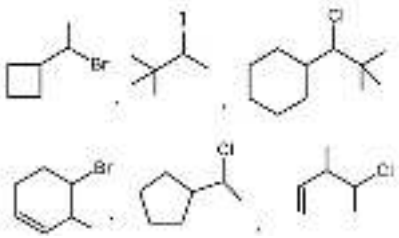
52) 

53) $CH_3–CH_2–CH_2–CH_2–OH \xrightarrow[S_N2]{NaBr, H_2SO_4} CH_3–CH_2–CH_2–CH_2–Br$

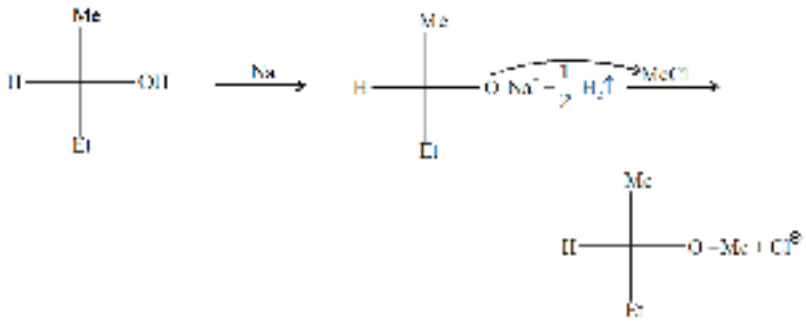
unbranched 1º alcohol give S_N2 reaction with HX.

54) 

In acidic medium S_N1 type product. $Nu^{\ominus}$ attack from crowded side.

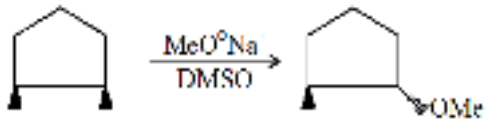
55) 

56)

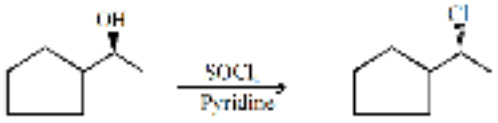


(A)

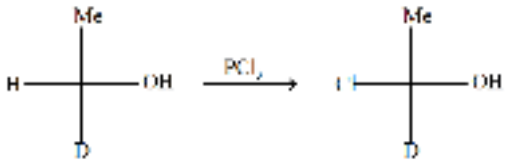
Retention of Product takes place

(B) 

SN^2 Mechanism will be performed

(C) 

Inversion

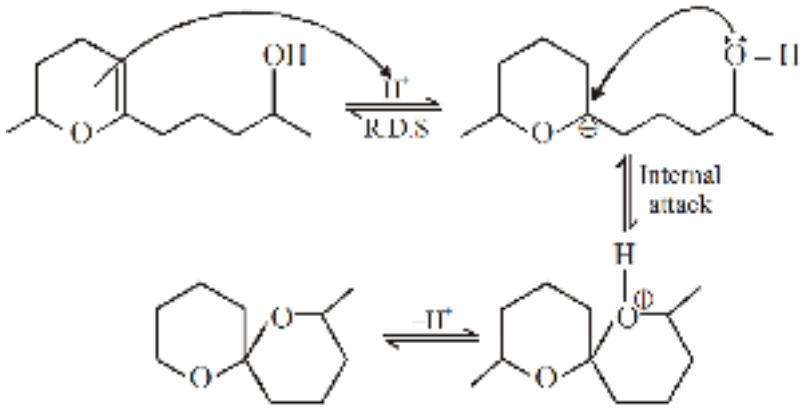
(D) 

Inversion

Only A option is correct

57)

Conceptual

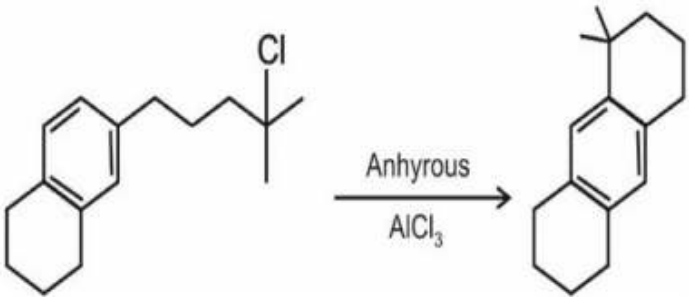
58) 

59)

Conceptual

60)

Conceptual

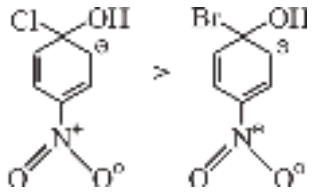
61) 

62)

Conceptual

63)

Conceptual

64)  Stability order If withdrawing groups are present at o, p position then rate of S_NAr increases.

65)

Conceptual

66)

Conceptual

67)

Conceptual

68)

Conceptual

69)

Conceptual

70)

Conceptual

71) S_{N^2} involves inversion of configuration

72)

Conceptual

73)

Conceptual

74)

Conceptual

75)

Conceptual

76)

Conceptual

77)

Conceptual

78)

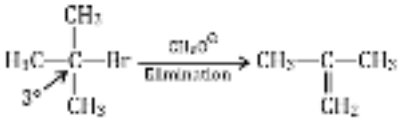
Conceptual

79)

NCERT-XII-II, Pg. # 215/216

80)

Conceptual

81) 

82)

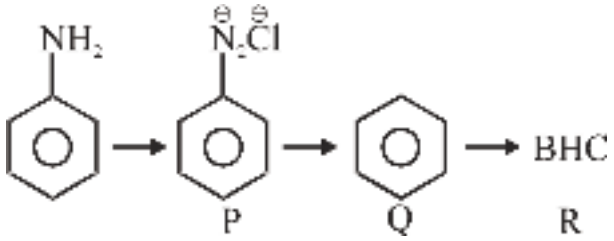
Conceptual

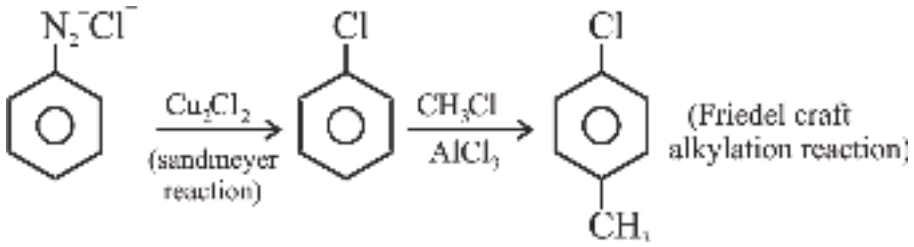
83)

Conceptual

84)

Conceptual

85) 

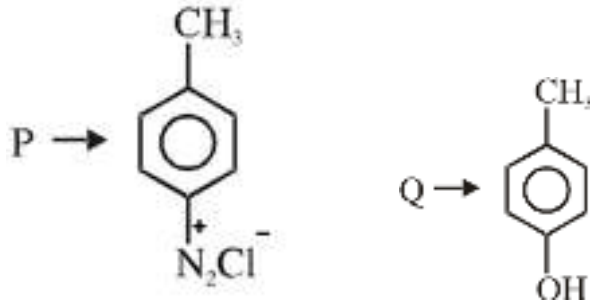
86) 

87)

Conceptual

88)

Conceptual

89) 

90)

Conceptual

BIOLOGY

91) A = (iv), B = (iii), C = (ii), D = (i)

92) Deoxyribonucleoside monophosphates

93) Sol; $\frac{5}{3}$

94) The backbone is constituted by sugar-phosphate and bases project inside

95) Positively charged polyamines (positively charged proteins)

96) 2(H2A + H2B + H3 + H4)

97) PDB = 6, NGB = 8

98) A = DNA, B = H1, C = Nu-body

99) A + C = G + T

100) It is single stranded DNA

101) **Statement I** is incorrect but **Statement II** is true.

102) (A), (C), (D) only

103) If both A and R are true but R is not the correct explanation of A then mark.

104) If both A and R are true and R is the correct explanation of A.

105) Mixing heat-killed pathogenic strain of bacteria with living non-pathogenic strain can convert some of the living cells into the pathogenic form

106) Avery, Leod and Carty, who proved the 10 transforming substance is DNA

107) LHS = Radioactivity detected in supernatant but not in cells
RHS = Radioactivity detected in cells but not in supernatant

108) Both Statement I and Statement II are true.

109) Both Statement I and Statement II are true.

110) All of the above

111) (iv) and (v)

112) (i), (ii), (iii) and (v)

113) If both A and R are true but R is not the correct explanation of A.

114) Adenosine, guanosine and cytidines are nucleotides.

115) Phosphodiester bonds

116) Only I, III and IV

117) Both statements are incorrect

118) Francis Crick

119) Sol : 24×10^6

120) H1 protein

121) A-Thymine; B-3.4 nm; C-Inside; D-2.2 m

122) Uniform Length throughout DNA

123) A - T; B - T; C - F; D - F

124) Nucleosomes in chromatin

125) Meselson and Stahl

126) It is single stranded DNA

127) It should not be able to express itself in the form of 'mendelian characters'

128) Difference in their lethality in mice

129) DNA synthesis

130) Alfred Hershey and Martha Chase

131) (A), (B), (C), and (D)

132) Only III and IV

133) Both (1) and (3)

134) a - 4, b - 3, c - 2,.d - 1

135) a - 4, b - 3, c - 1, d - 2

136) NCERT (XIIth) Pg. # 160 Para 8.5

137) NCERT Pg. No. 144

138) Ncert Pg No: 142, 143

139) NCERT Pg. # 143

140) NCERT Pg. # 143

141)

NCERT Page No.158,159,160

142) Ncert Page 142 and 143.

143) Ncert Page 138.

144)

NCERT XII, Pg # 160

145) NCERT XII Pg.#158 (E) 171 (H)

146)

Generated by Allie

Problem Statement: The question asks to identify which statement about smoking is incorrect among a given list of statements related to its health effects and physiological impact.

Underlying Concept: The core concept involves understanding the health effects of smoking, particularly how chemicals in tobacco smoke affect the body systems such as the

cardiovascular and respiratory systems, as well as associations with diseases. It also touches on the biochemical effects of nicotine and carbon monoxide on the blood and adrenal glands.

Tips and Tricks: Recall the direct effects of smoking on lungs, heart, and blood content, and differentiate these from diseases mainly caused by other risk factors.

Common Mistakes: Confusing smoking-related diseases with those caused primarily by other factors like alcohol, leading to incorrect association of smoking with liver cirrhosis.

Why Other Options Are Incorrect?: Although smoking has widespread health effects, it is not directly associated with causing liver cirrhosis. The other statements correctly describe known physiological impacts of smoking such as increased carbon monoxide in blood and nicotine-induced adrenaline release.

147) NCERT-XII, Pg. # 158-159

148)

Anabolic steroids are synthetic compounds similar to the male sex hormone testosterone. Use of anabolic steroids in females produces such as masculinization, excessive hair growth on face and body.

149) NCERT (XIIth) Pg. # 158,159

150) NCERT XII Pg. # 161

151) Statement I: Correct. The text clearly indicates that the addictive potential of drugs and alcohol can lead to a vicious cycle of regular use and abuse.
Statement II: Incorrect. Dependence is characterized by an unpleasant withdrawal syndrome, not a pleasant one, when the regular dose of drugs or alcohol is abruptly discontinued.

152) NCERT XII Pg. # 161

153) NCERT Pg No: 160

154) NCERT-XIIth, Pg. # 158,159

155) NCERT, Pg. # 142-143

156) NCERT Pg. No. 169

157)

NCERT Pg. No. 178

158) NCERT (XIIth) Pg. # 196,199

159) NCERT (XII) Pg. # 201, para-11.3

160)

The correct answer is **option 4: P-iii, Q-i, R-ii, S-iv**
Explanation:

P. DNA Ligase: This enzyme connects DNA fragments by creating phosphodiester bonds. Therefore, P corresponds with iii.

Q. Restriction Exonuclease: Exonucleases work by removing nucleotides from the ends of DNA strands. Thus, Q aligns with i.

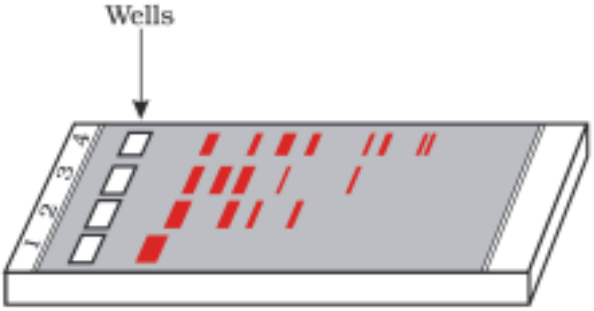
R. Taq Polymerase: Taq polymerase is a type of DNA polymerase utilized in PCR. It adds nucleotides to extend the primer on a DNA template. Hence, R corresponds with ii.

S. Restriction Endonuclease: Endonucleases are responsible for cutting DNA at specific internal sites known as restriction sites. Therefore, S matches with iv.

161) NCERT Pg. # 201

162)

Explaining the Question :-
The question is asking about Match the following
Concept:-
AGROSE GEL ELECTROPHORESIS
Solution :-



Well-1 → Undigested DNA (largest, near well)
Well-2 → Medium restriction (moderate cuts)
Well-4 → Maximum restriction site (many small fragments)
Smallest DNA → Closest to anode (moves farthest)
Largest DNA → Closest to cathode (moves least)

163)

The answer is:2. Two.
You need two antibiotic resistance genes on a plasmid for successful ligation. One gene selects for cells that took up the plasmid, and the second helps identify cells where the alien DNA was inserted correctly (often by disrupting that second gene).

164)

The correct answer is: 3. Both statements are correct.
Both statements are correct. EcoRI cuts DNA at the GAATTC sequence, specifically between G and A, creating staggered cuts (sticky ends).

165)

Explanation:

Transformed cells (with ampicillin resistance gene): Grow.

Untransformed cells (without gene): Die.

Answer: Option 3 is correct.

166) **Question Explanation :**
The question is asking about Fill in the blanks

Concept :
This question is based on Tools of RDT

Solution :
A: Same restriction enzyme cuts produce sticky ends, joined by DNA ligase.
B: Cell walls broken by lysozyme (bacteria), cellulase (plants), chitinase (fungi).
C: Protein made in a different host = recombinant protein.
D: DNA stained with ethidium bromide and seen under UV light.

Final Answer :
Option (4)

167) **Question Explanation :**
The question is asking about Among many, the two core techniques that enabled the birth of modern biotechnology are

Concept :
This question is based on Genetic engineering and Bioprocess engineering

Solution :
Modern biotechnology emerged mainly from genetic engineering (manipulating DNA) and bioprocess engineering (large-scale production using biological systems).Correct Answer: (2) Genetic engineering and Bioprocess engineering

Final Answer :

Option (2)

168) **Question Explanation :**
The question is asking about a question based on your assertion-reason format:

Concept :
This question is based on Plasmids

Solution :
Assertion: Plasmids can be used as vectors to deliver alien DNA into host organisms because they carry genetic material that can be introduced into cells.
Reason: Plasmids are autonomously replicating circular extrachromosomal DNA, meaning they replicate independently of the host's chromosomal DNA, ensuring the alien DNA is maintained and expressed in the host.
Since both statements are true and the reason correctly explains the assertion, option (1) is the correct answer.
Both assertion and reason are true, and the reason correctly explains the assertion because plasmids replicate independently, making them ideal vectors for DNA delivery.Correct option: (1)

Final Answer :
Option (1)

169) The answer is **option 4. A - EcoR I, B - BamH I, C - *Amp*^{R}, D - ori**

170)

- The total size of the DNA is 12 kb.
- Two recognition sites for BamHI are present 4 kb apart from each other.
- One of the recognition sites is at 4 kb distance from one end.

When BamHI cuts the DNA, it will make two cuts at the recognition sites, which will result in three fragments:

A. The first fragment will be from the 5' end of the DNA to the first BamHI recognition site (4 kb).
B. The second fragment will be from the first BamHI site to the second BamHI site (4 kb).
C. The third fragment will be from the second BamHI site to the 3' end of the DNA (4 kb).

Since the DNA is completely digested, there will be **three fragments** formed.

171)

NCERT Page No. # 198, 199, 200, 195

172) NCERT Page No. # 195/200/204

173) **Question Explanation :**

The question is asking about two statements related to the origin of recombinant DNA (r-DNA) technology and identify the correct combination based on scientific facts

Concept :
This question is based on Recombinant DNA Technology -

Solution :
Statement-I: Incorrect
→ First r-DNA used E. coli plasmids, not Agrobacterium tumefaciens.
Statement-II: Incorrect
→ Cohen and Boyer used antibiotic resistance genes, not vitamin A genes.

Hence,
Option (2) is correct

174) **Question Explanation :**
The question is asking about how many of the listed elements are integrated in biotechnology for products and services.

Concept :
This question is based on Concept of Biotechnology Integration.

Solution :
Biotechnology integrates all five elements: natural science, organisms, cells, parts of cells, and molecular analogues to develop products and services.

Hence,
Option (1) is correct : Five

175) NCERT page- 167, 168

176) NCERT Page no 199

177) NCERT page- 166, 167

178) **Explaining the Question :-**The question is asking about which of the following statements about DNA elution, plasmid pBR322, downstream processing, and bacterial competence are correct?
Concept:- Key Terms in Genetic Engineering and Biotechnology
]
Solution :-All given statements (a), (b), (c) and (d) are correct regarding DNA elution, pBR322 features, downstream processing, and bacterial
competence.
Final Answer :- (4) All are correct

179)

NCERT-XII, Page # 170

180) NCERT, Pg. # 168 & 172